

I: Image Formation

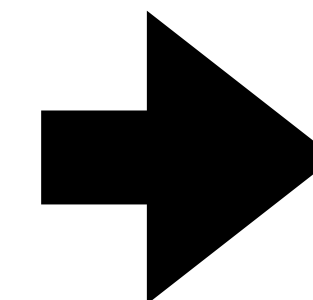
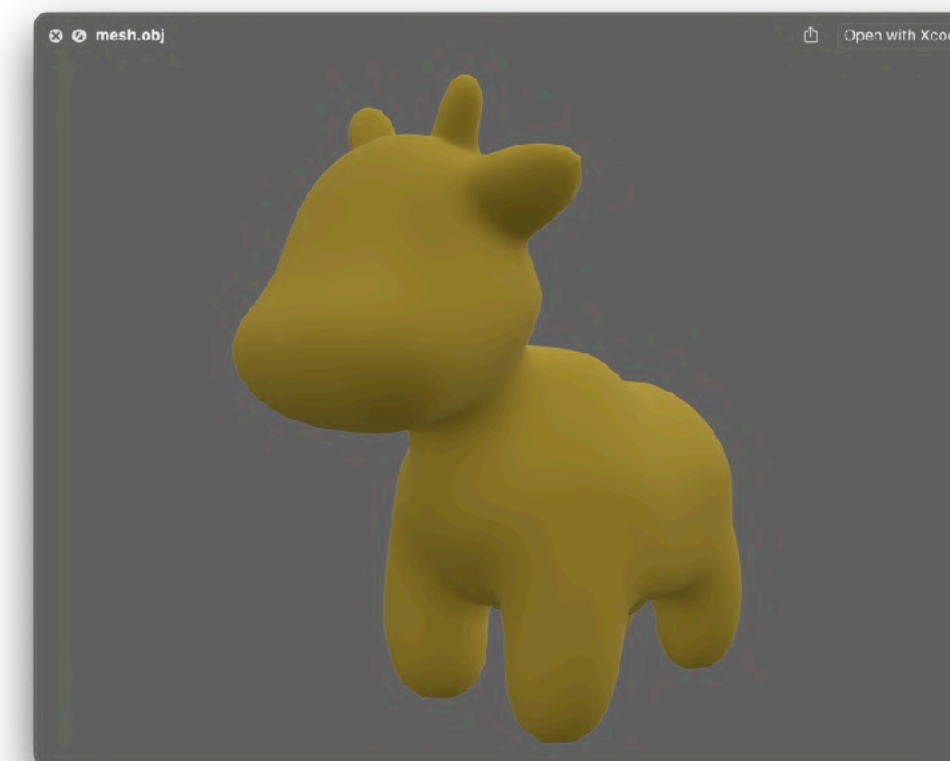
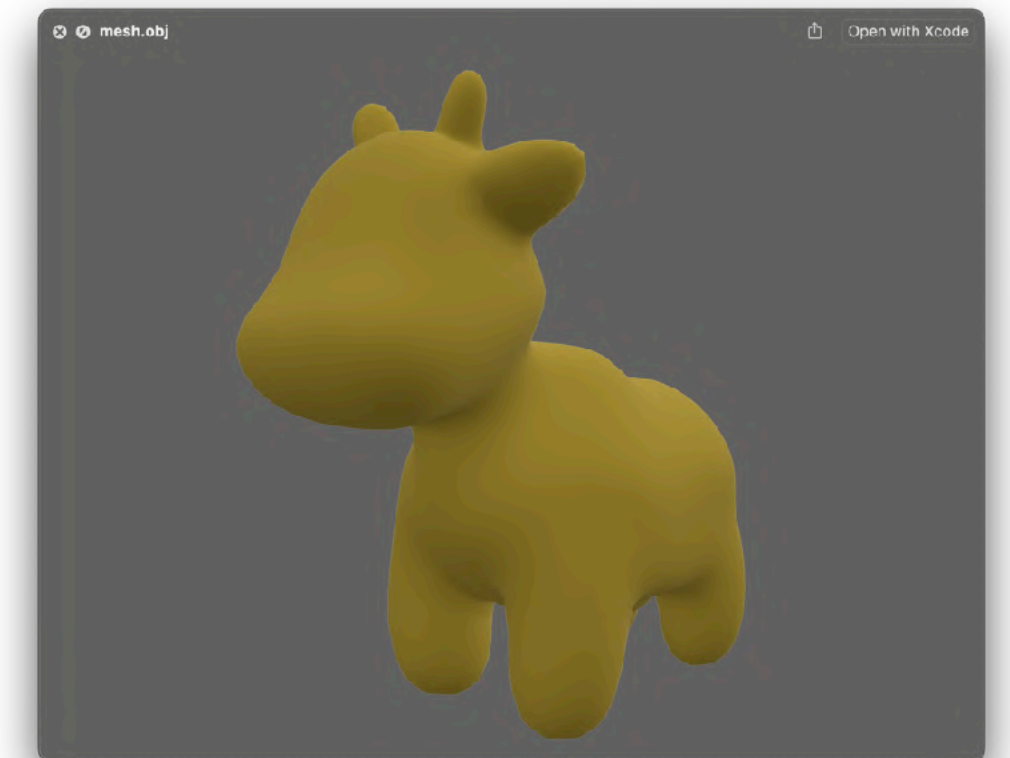
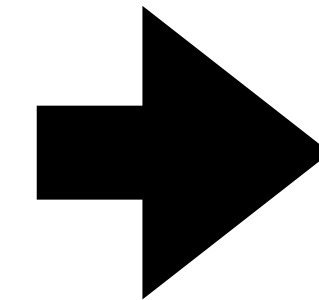
3D CV

Kirill Struminsky, Mishan Aliev

3D CV

- Graphics
 - Get image from description
- Computer vision
 - Get description of an image
- 3D Computer vision
 - What is the shape of the cow?

```
mtllib mesh.mtl
g default
v 0.3908158838748932 0.5781099796295166 -0.5622197389602661
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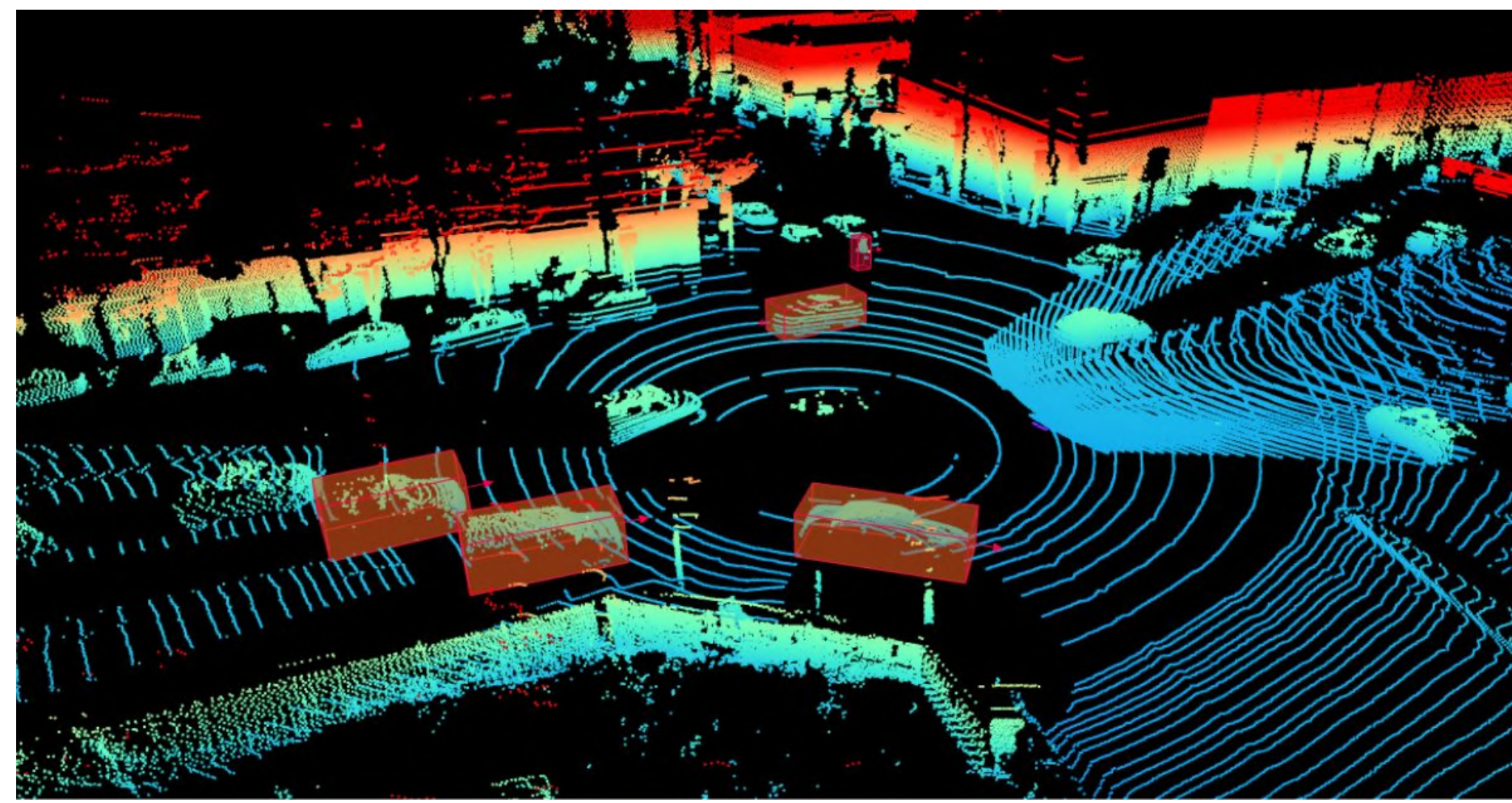
“A cow”

Why?

Navigation in the Real World



Zero DoF



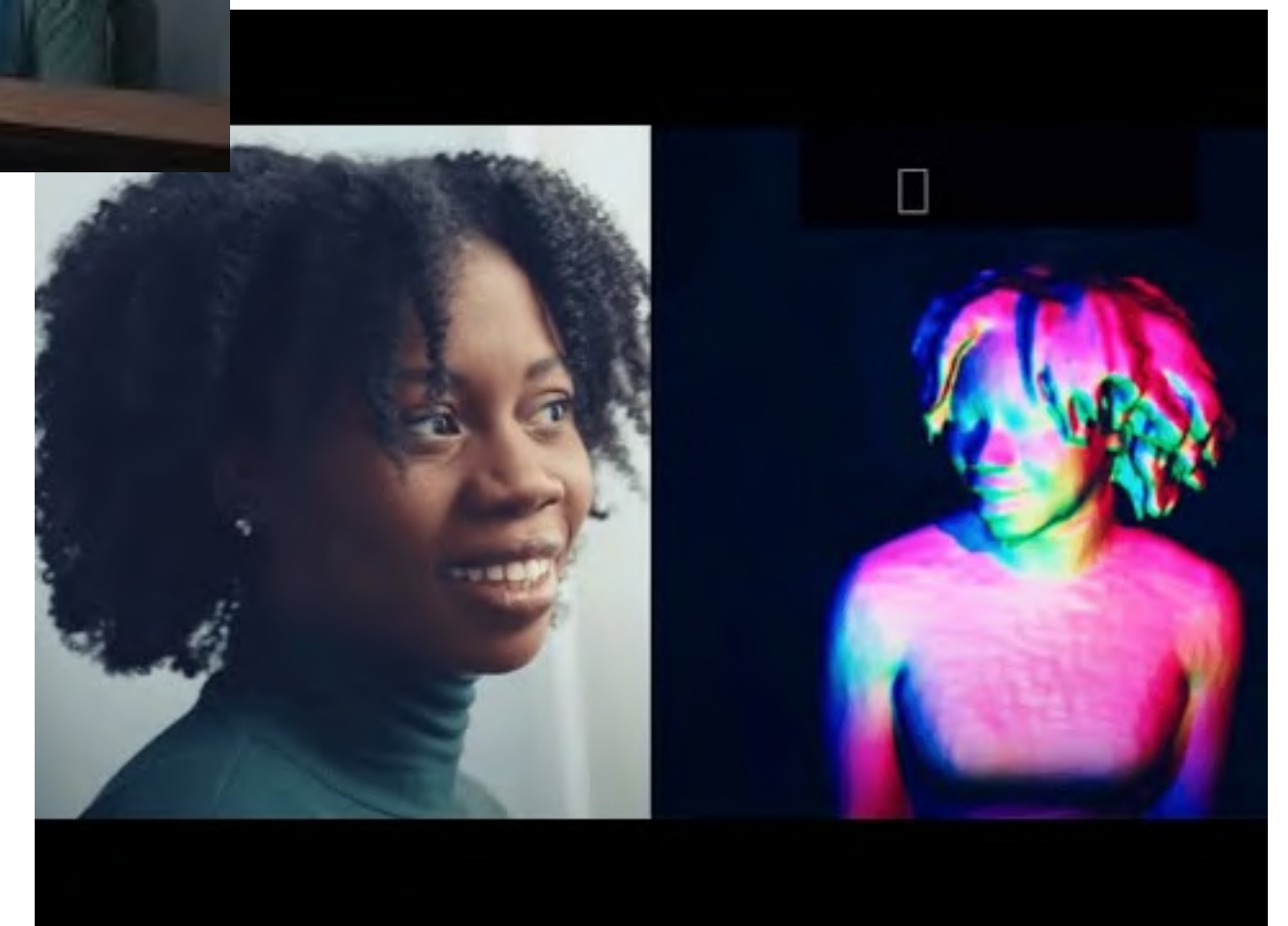
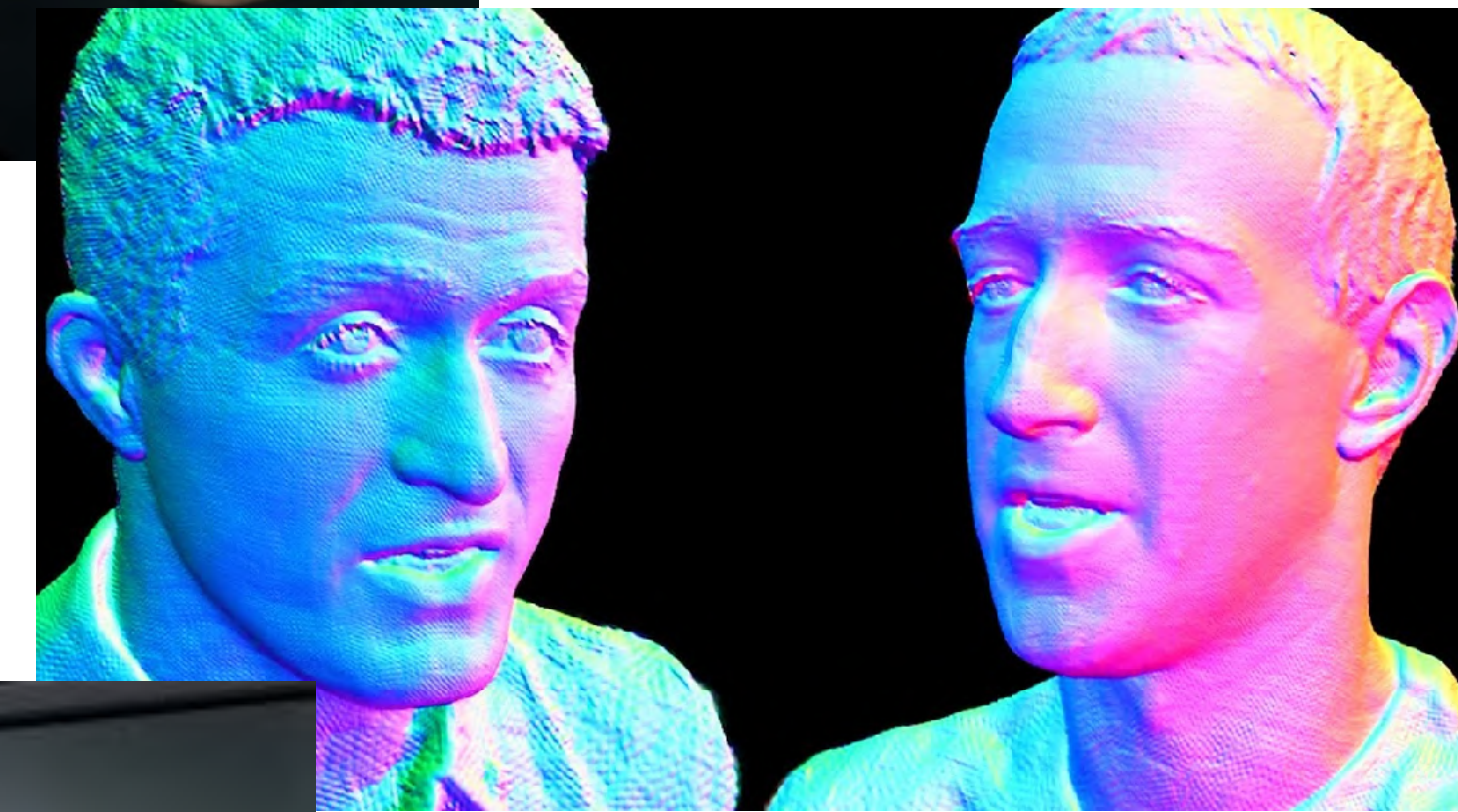
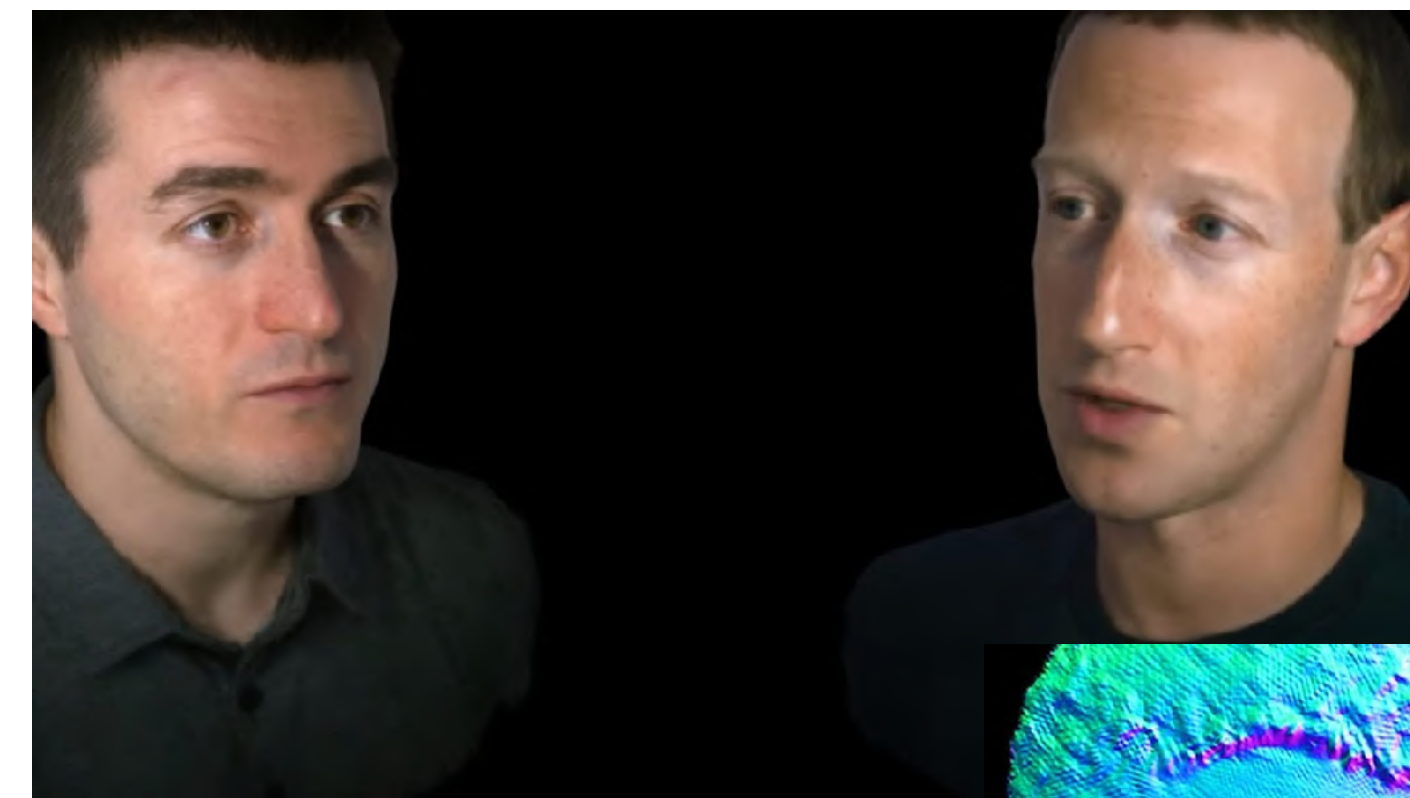
Several DoF



Many DoF

Why?

Graphics, AR & VR



Course Overview: Lecture 1 + 2

Classic Multi-View Geometry



Course Overview: Lecture 3 + 4

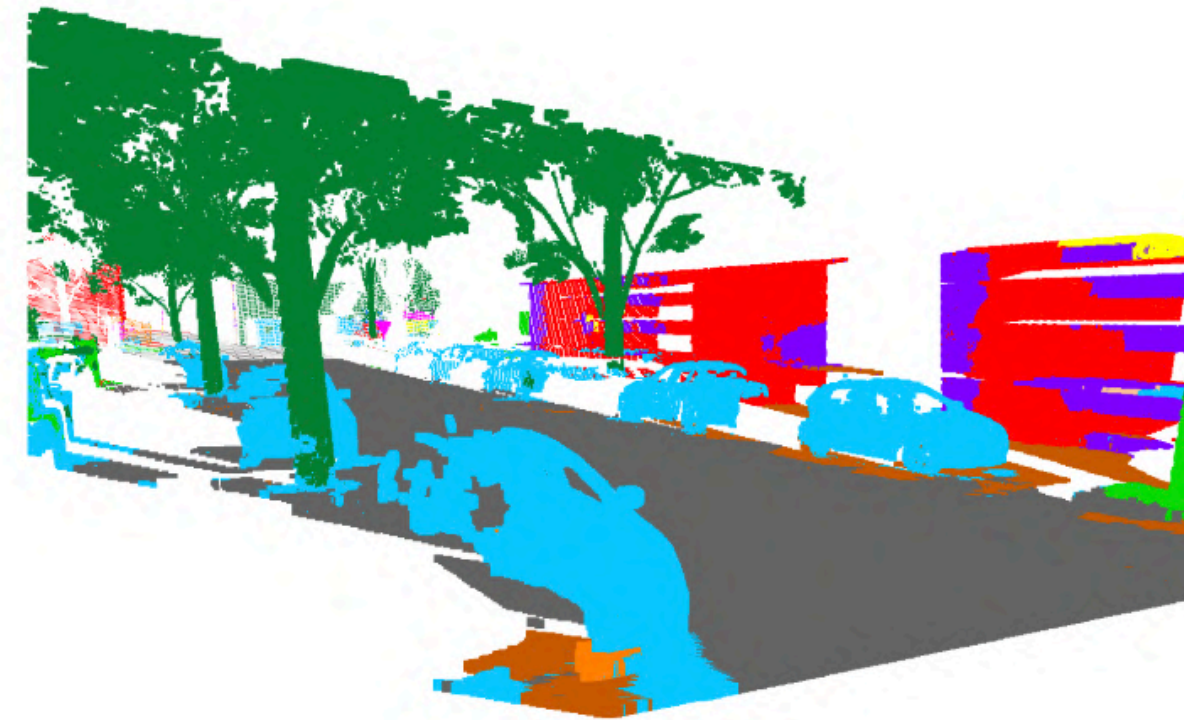
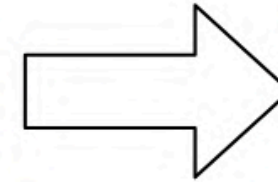
ML for Multi-View Geometry & Depth Estimation (+ Assignment)



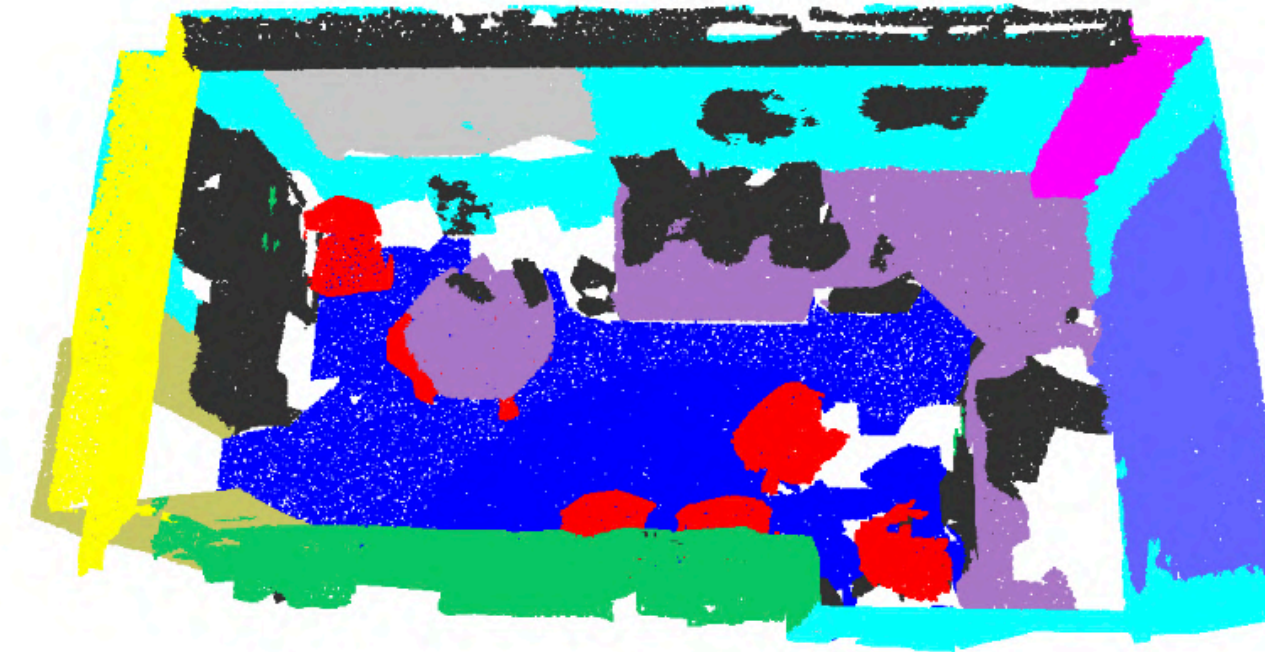
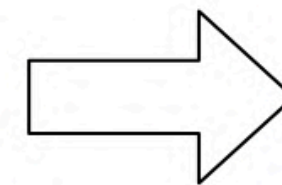
Course Overview: Lecture 5

Point-Cloud Processing (+ Assignment)

Outdoor Scene



Indoor Scene

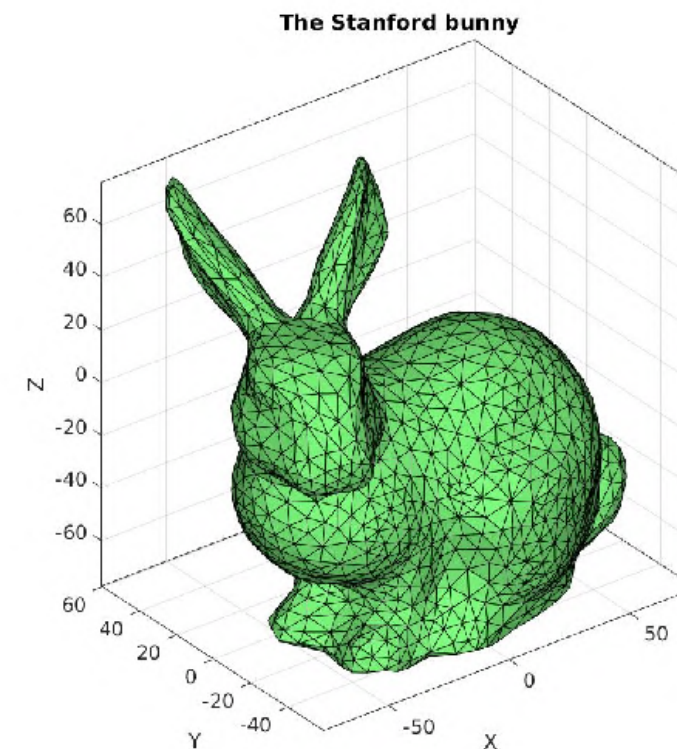


Input: 3D Point Cloud

Output: Semantic Segmentation

Course Overview: Lectures 6-7

Classic Graphics Pipeline, Differentiable Rasterization, Meshes



(a) $\bar{\mathbf{T}}, \mathcal{W}$



(b) $\bar{\mathbf{T}} + B_S(\vec{\beta}), J(\vec{\beta})$



(c) $T_P(\vec{\beta}, \vec{\theta}) = \bar{\mathbf{T}} + B_S(\vec{\beta}) + B_P(\vec{\theta})$



(d) $W(T_P(\vec{\beta}, \vec{\theta}), J(\vec{\beta}), \vec{\theta}, \mathcal{W})$

Course Overview: Lectures 8-11

Radiance Fields, Differentiable Rendering for Reconstruction & Generation (+ Assignment)

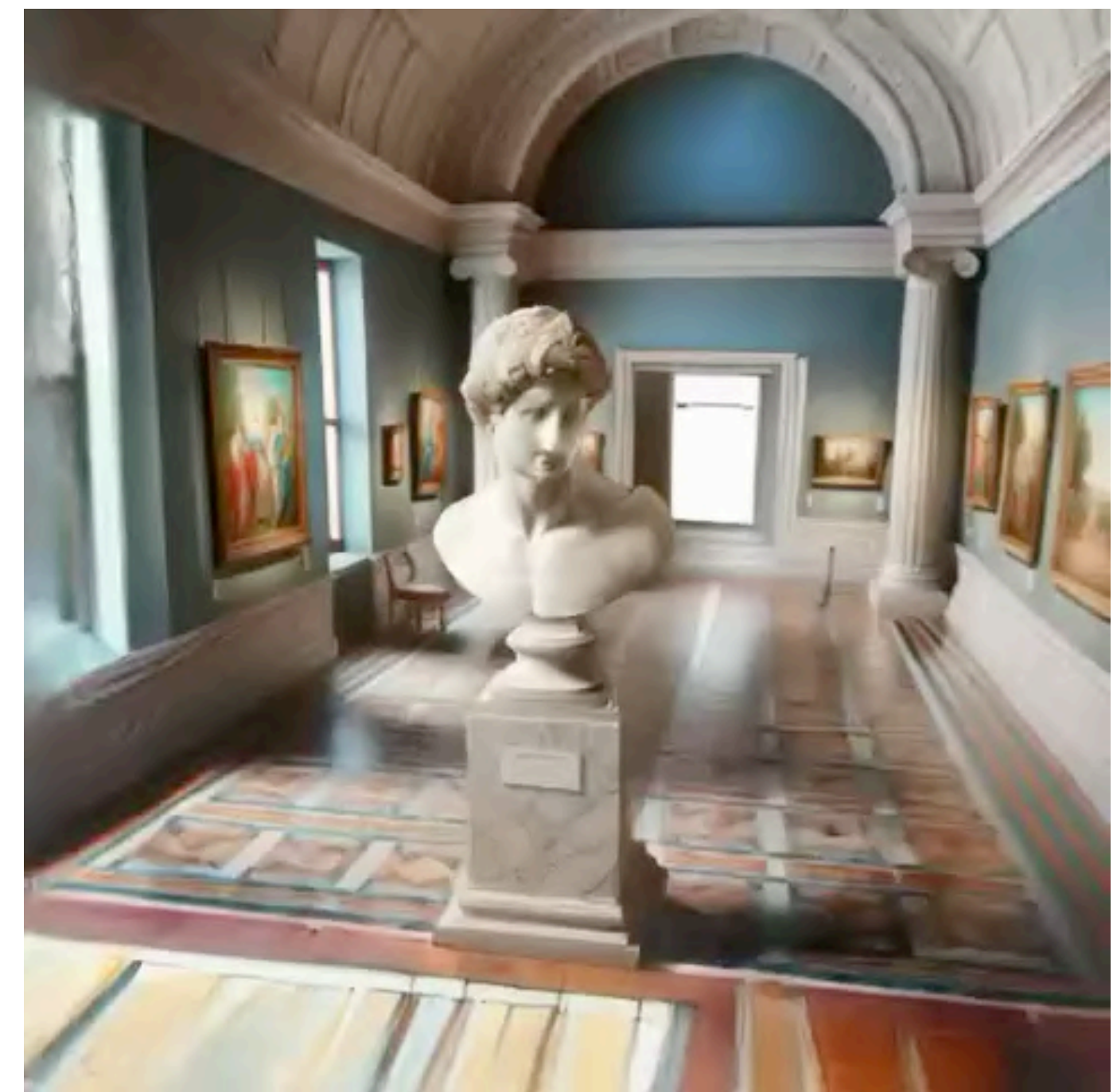
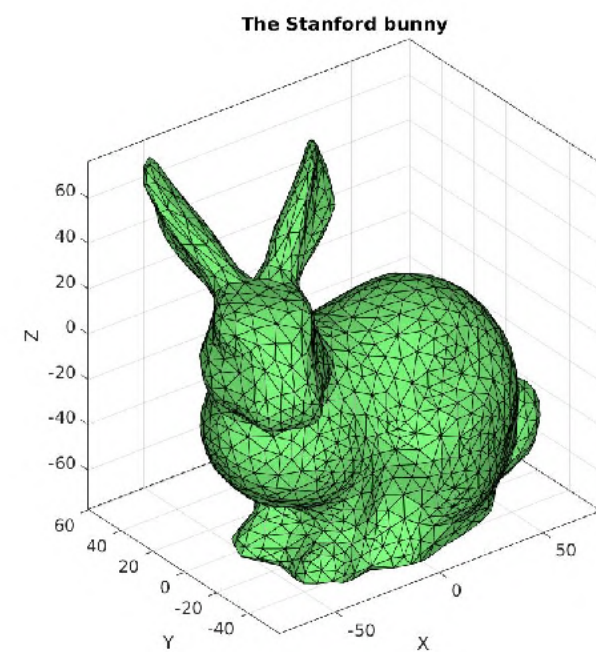
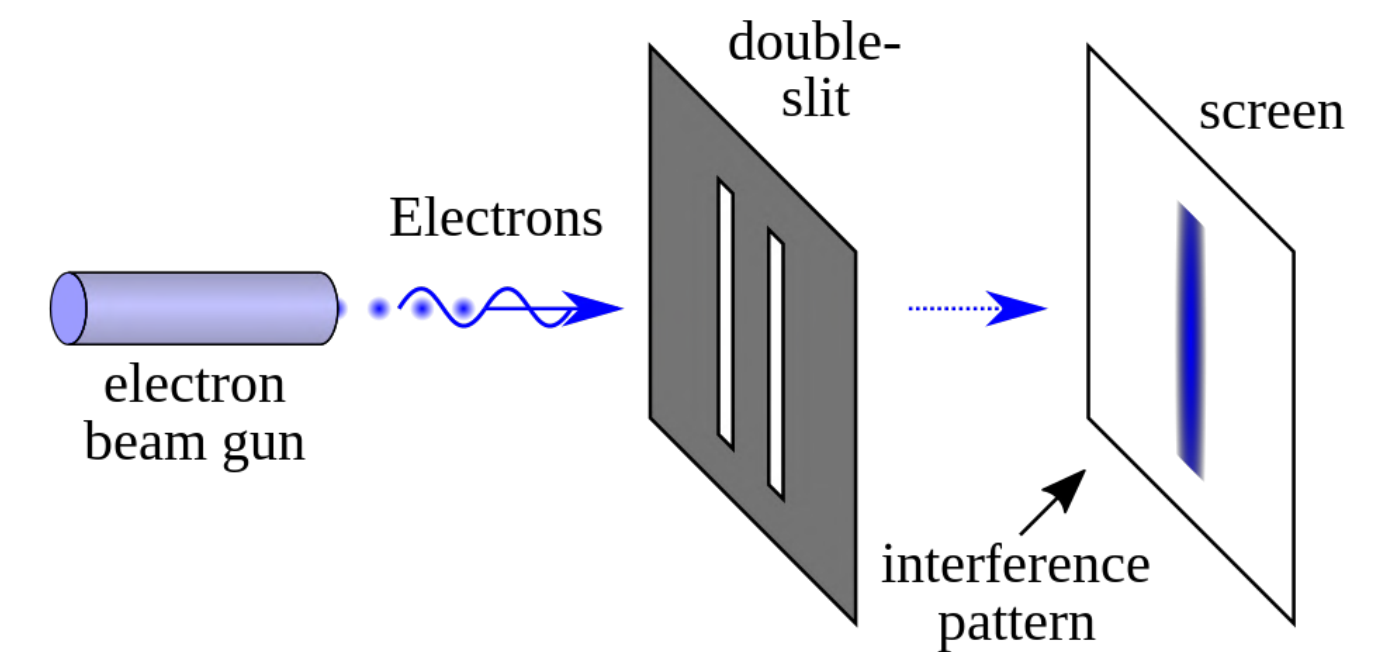
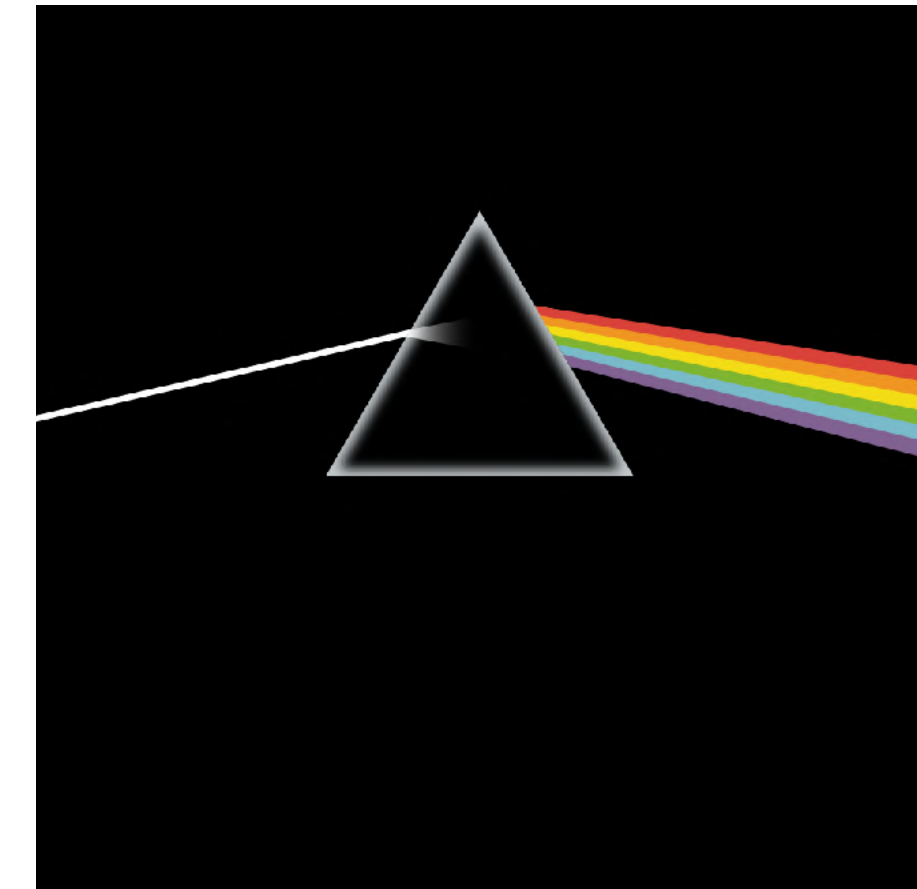


Image Formation

Light

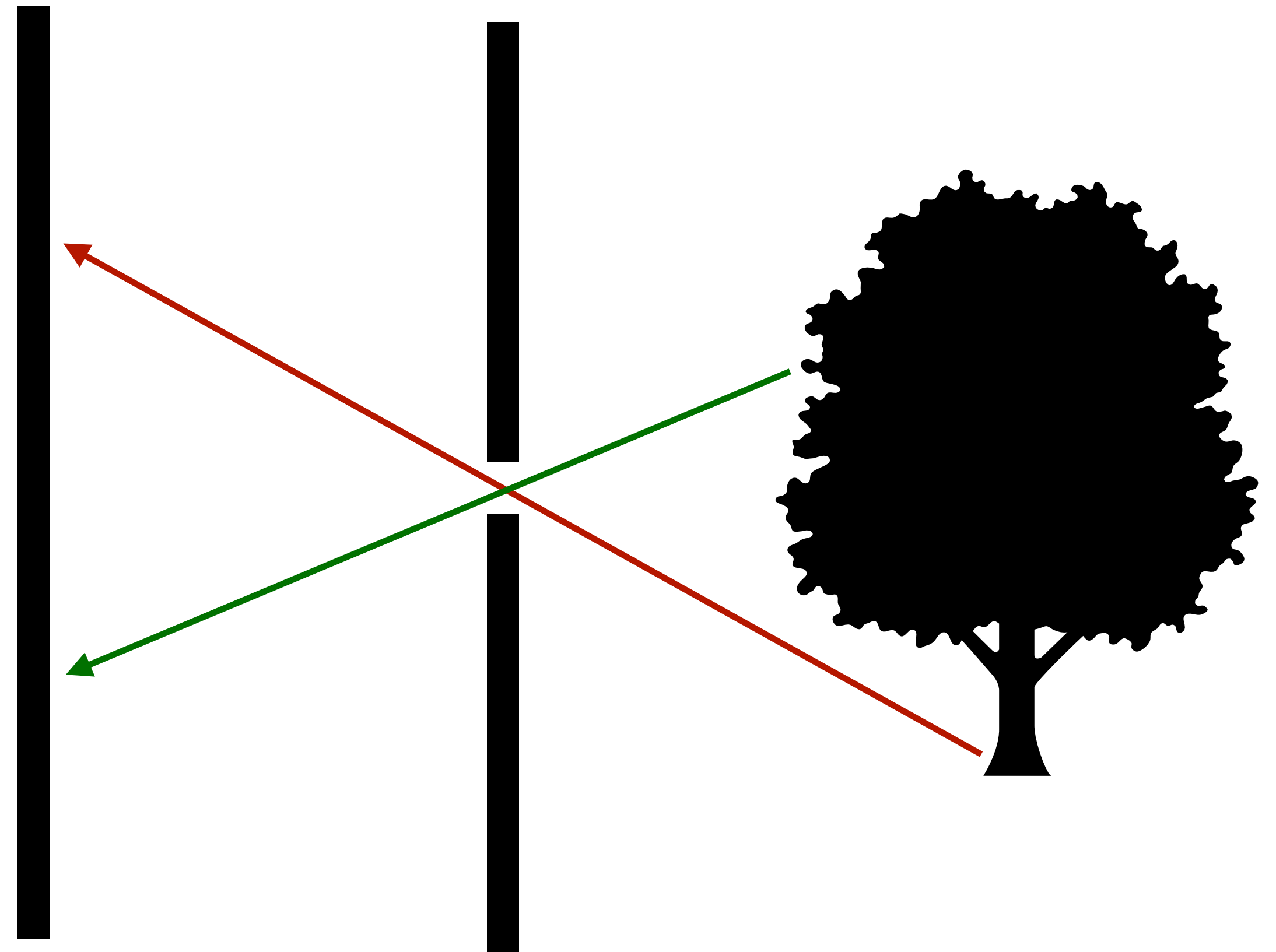
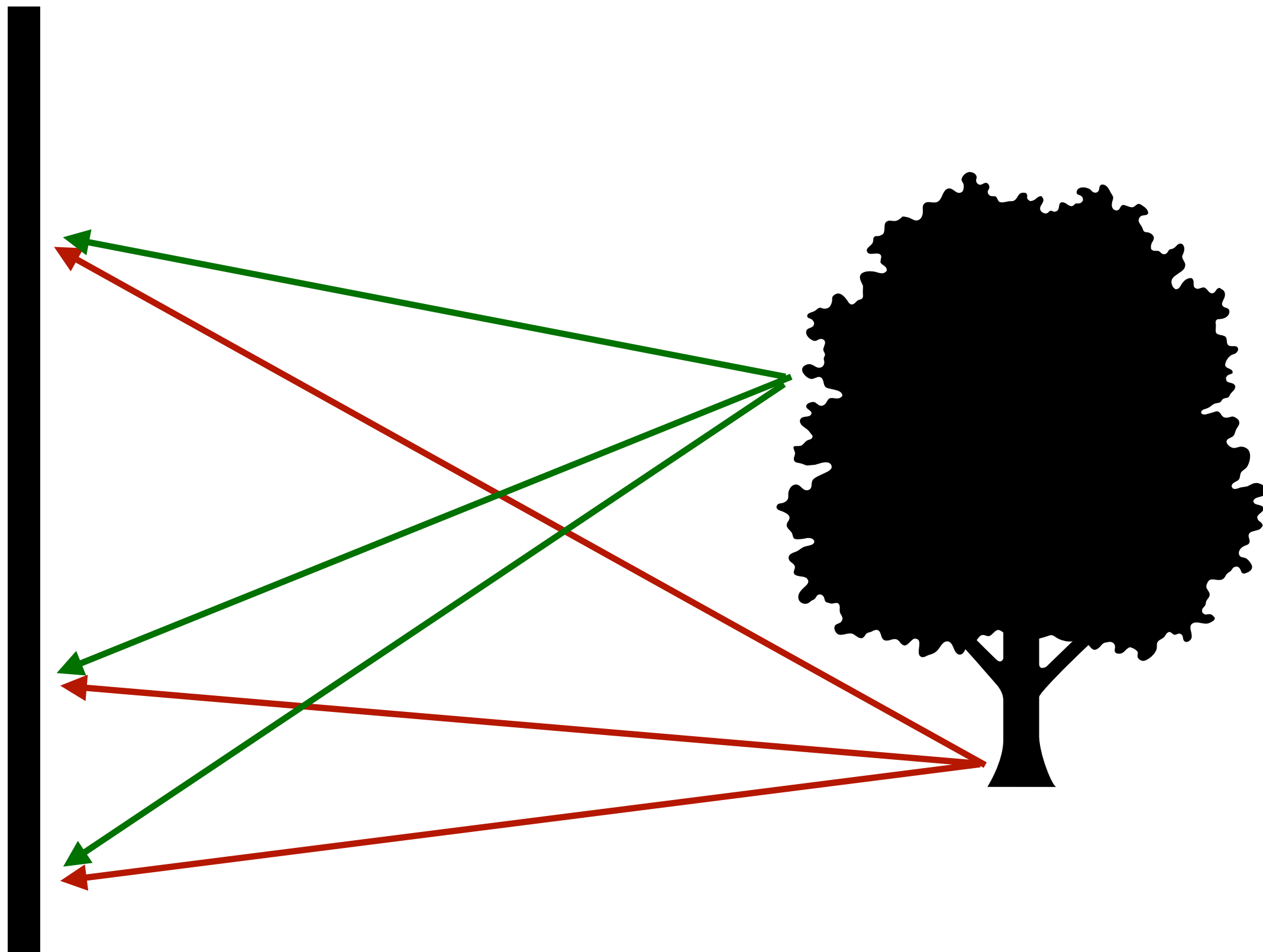


https://en.wikipedia.org/wiki/Double-slit_experiment

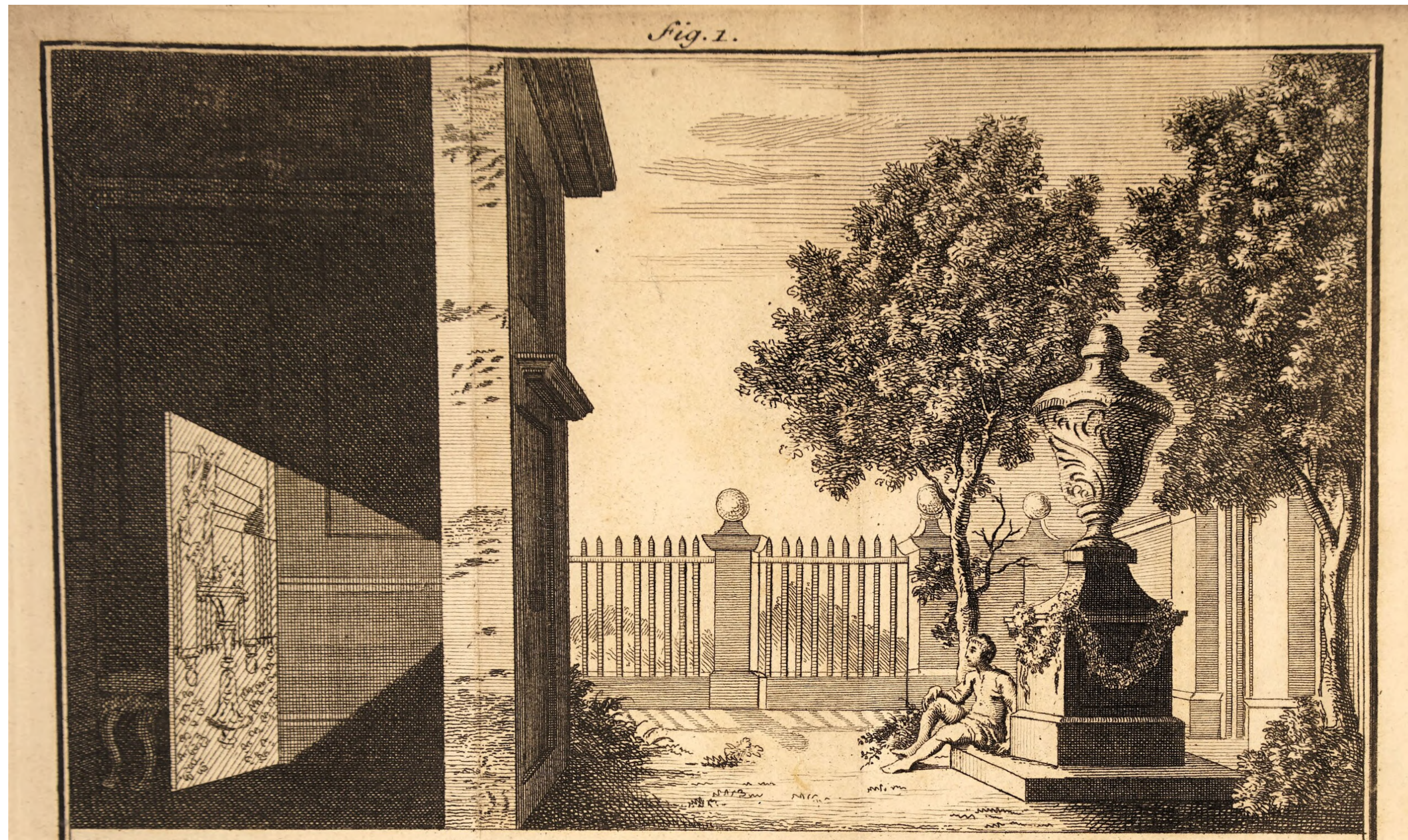
https://ru.wikipedia.org/wiki/The_Dark_Side_of_the_Moon

https://www.youtube.com/watch?v=frLwRLS_ZR0

Simplest Camera



Camera Obscura and Pinhole Camera



Natural Camera Obscura

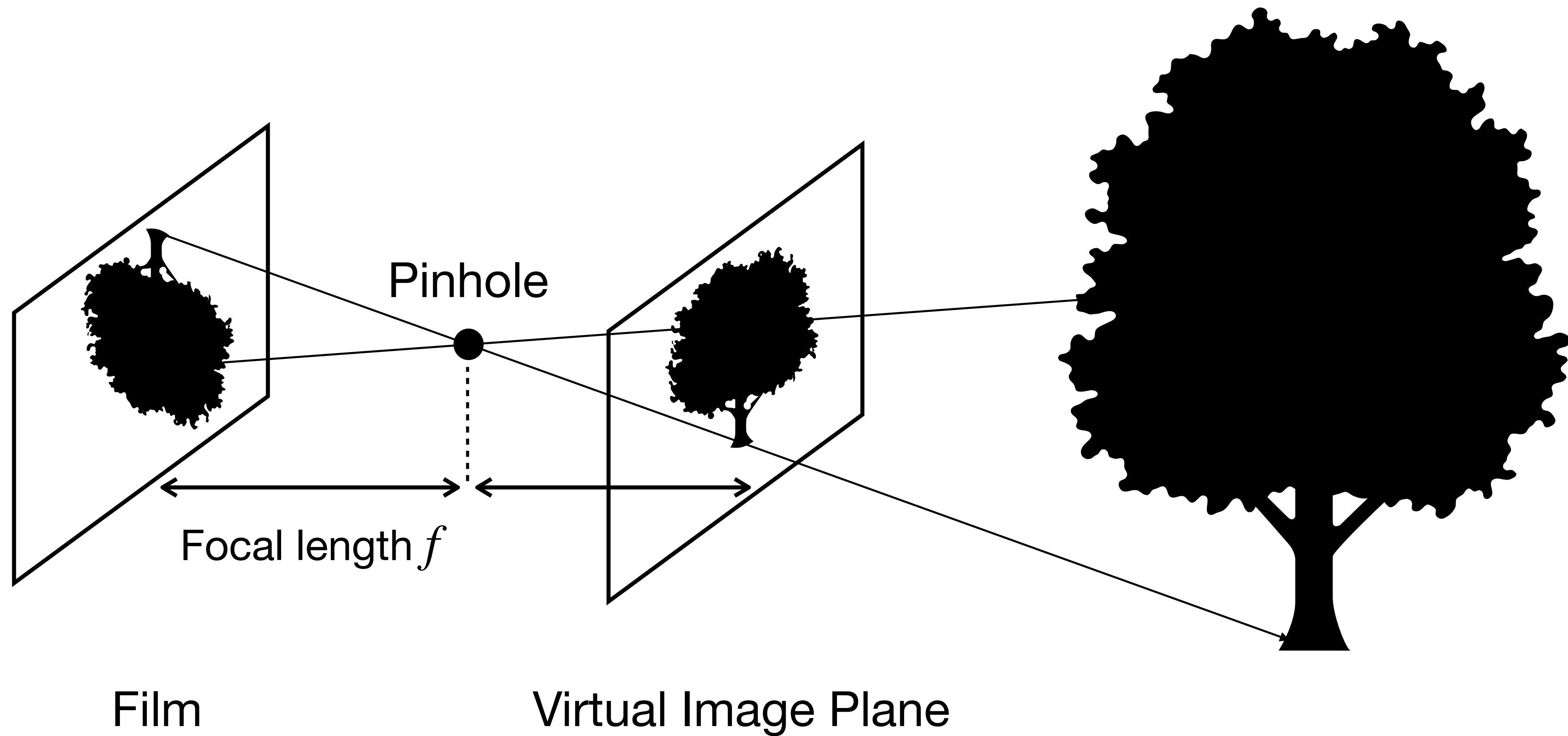


https://en.wikipedia.org/wiki/Camera_obscura

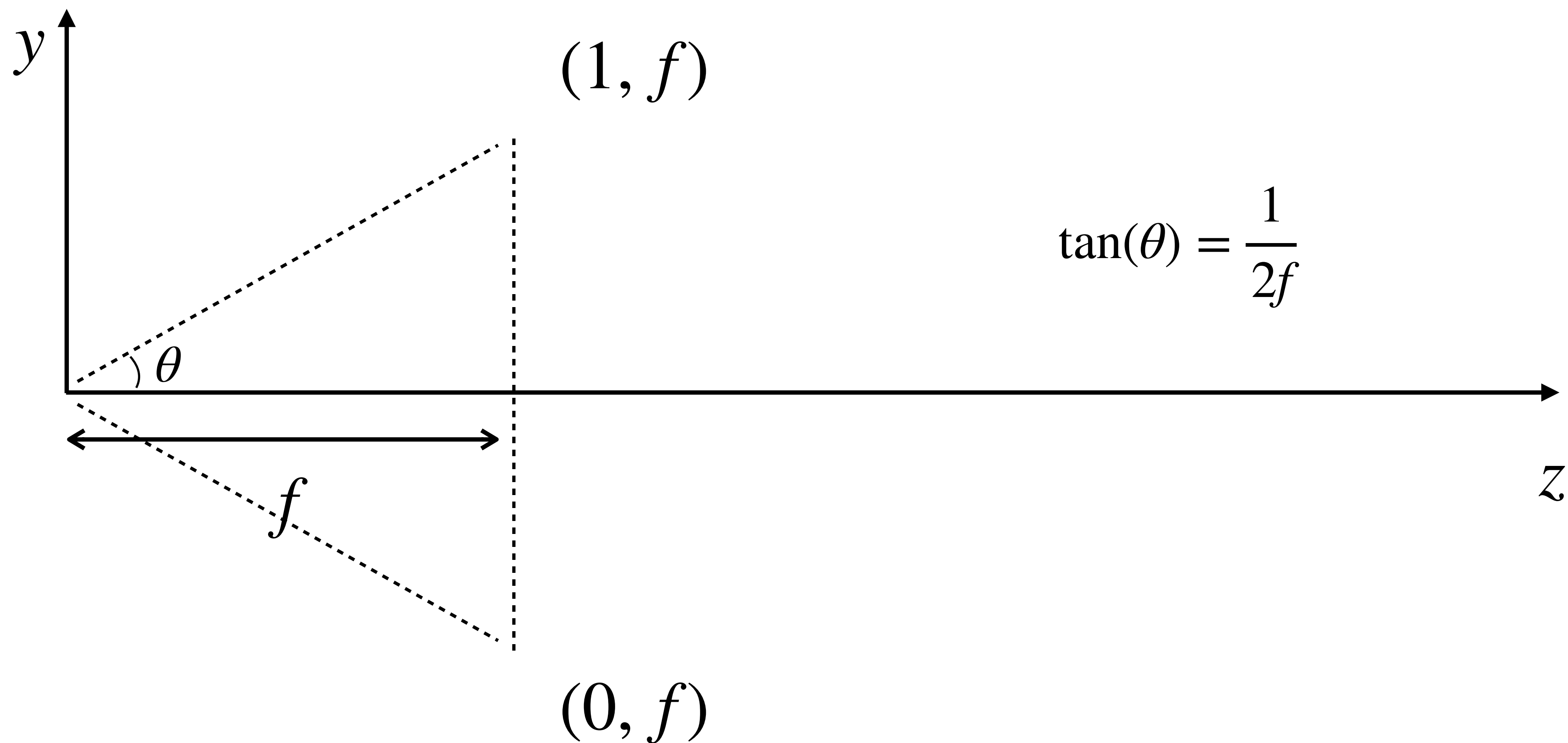
https://www.reddit.com/r/pics/comments/3gplnx/sunlight_through_a_tree_during_an_eclipse/

see also <https://www.abelardomorell.net/selectedworks/camera-obscura>

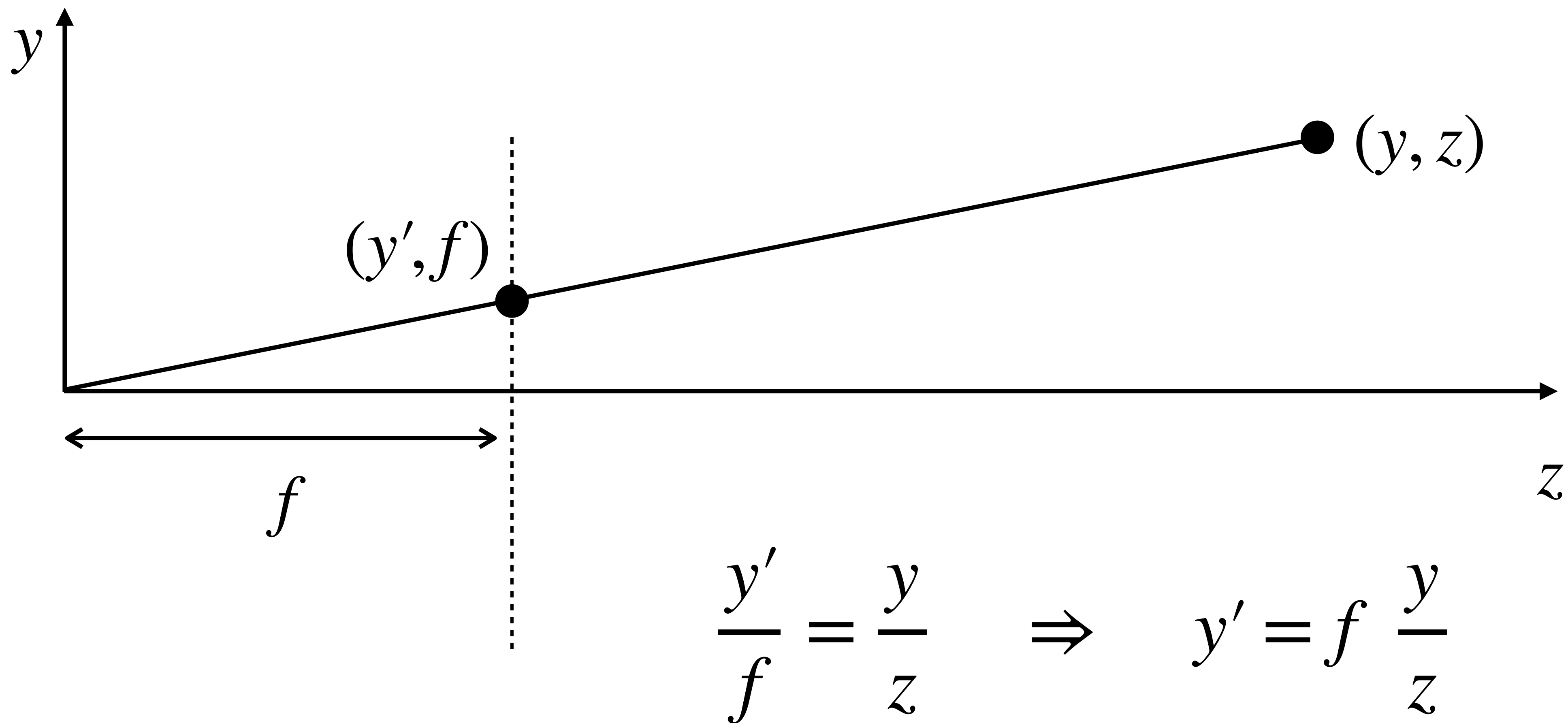
Perspective Projection



Field of View vs Focal Length



Perspective Projection



Homogeneous Coordinates

Perspective Projection as a Linear Map

Inhomogeneous coordinates:

Projection is a non-linear map: $(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$

Homogeneous coordinates:

Take $X = (x, y, z)$ and introduce an additional coordinate (homogeneous coordinates):

$$\tilde{X} = (x, y, z, 1)$$

We say $\tilde{Y} = \tilde{X}$ if $\exists \lambda \in \mathbb{R}$ s.t. $\tilde{Y} = (\lambda x, \lambda y, \lambda z, \lambda)$

Projection turns into $(x, y, z, w) \rightarrow (x, y, \frac{z}{f})$

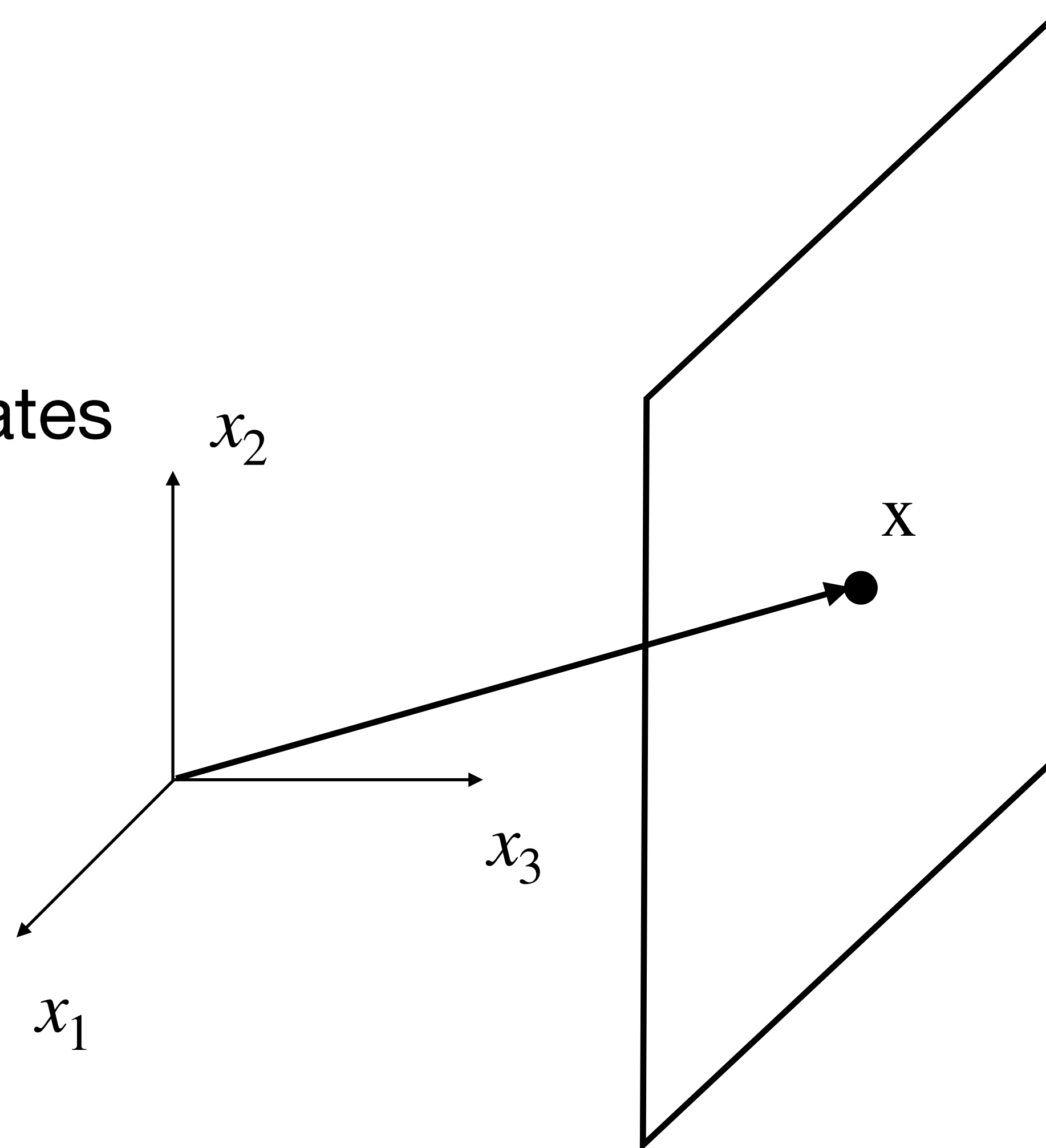
$$H = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{f} & 0 \end{pmatrix}$$

A Glimpse into Projective Geometry

A model of the Projective Plane

Homogeneous Coordinates

$$\mathbf{x} = (\lambda x_1 : \lambda x_2 : \lambda)$$



2D Coordinates

$$\mathbf{x} = (x_1, x_2)$$

A Glimpse into Projective Geometry

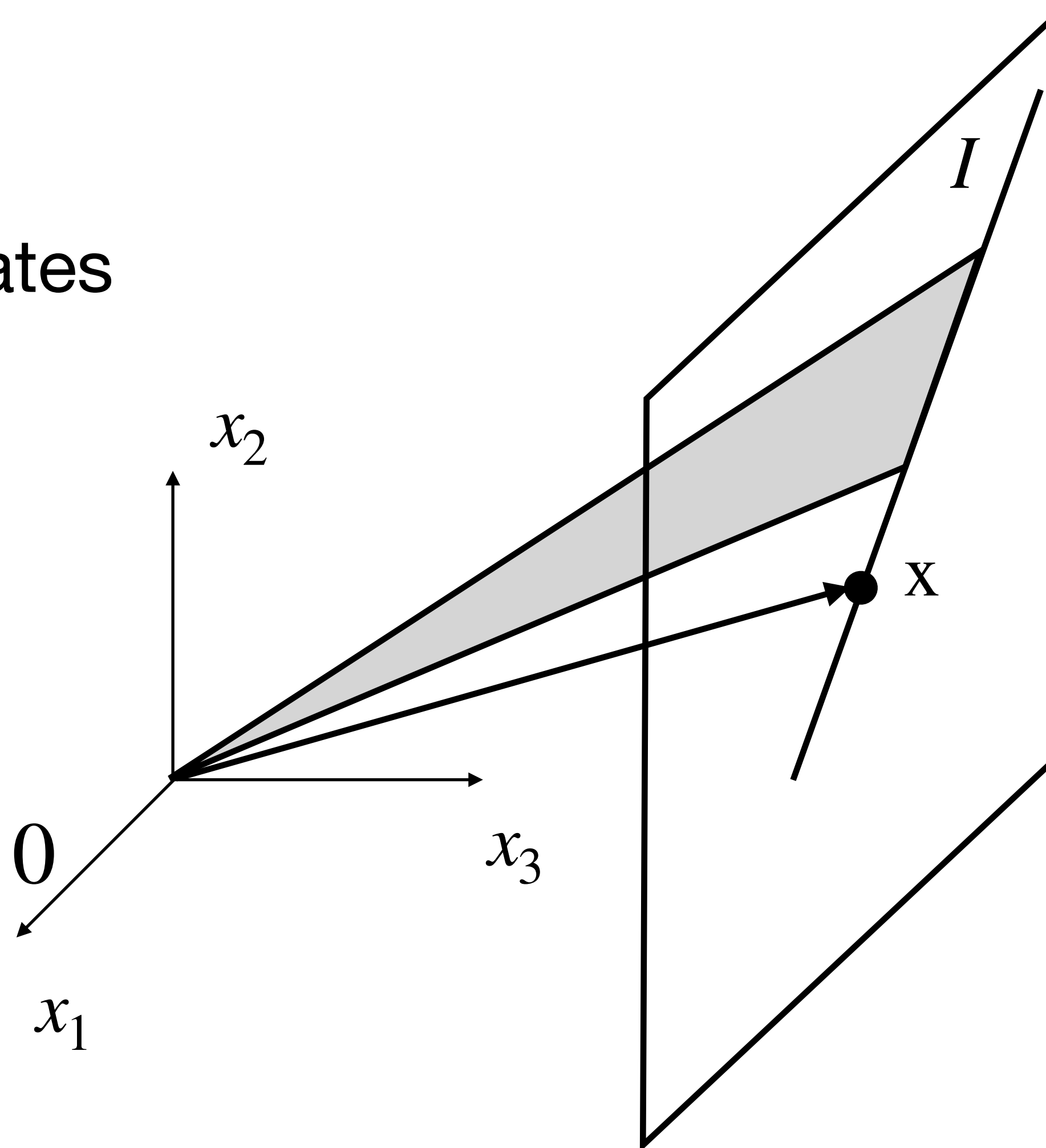
A model of the Projective Plane

Homogeneous Coordinates

$$\mathbf{x} = (\lambda x_1 : \lambda x_2 : \lambda)$$

Line:

$$I : ax_1 + bx_2 + x_3c = 0$$



2D Coordinates

$$\mathbf{x} = (x_1, x_2)$$

Line:

$$I : ax_1 + bx_2 + c = 0$$

A Glimpse into Projective Geometry

A model of the Projective Plane

Projective space $\mathbb{P}^n$ is the set of equivalence classes of $\mathbb{R}^{n+1} \setminus \{0\}$ under relation $\sim$, where $x \sim y$ if $\exists \lambda : x = \lambda y$

Linear map $\mathbb{R}^{n+1} \rightarrow \mathbb{R}^{m+1}$ induce transformations of projective space

Example: simple projection

$$H = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



Space Transformations

Transformation:

Translation

$$T = \begin{pmatrix} 1 & 0 & 0 & t_1 \\ 0 & 1 & 0 & t_2 \\ 0 & 0 & 1 & t_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Rigid Movement

$$R = \begin{pmatrix} Q & t \\ 0 & 1 \end{pmatrix}$$

Similarity

$$S = \begin{pmatrix} \lambda Q & t \\ 0 & 1 \end{pmatrix}$$

Affine Transformation

$$S = \begin{pmatrix} A & t \\ 0 & 1 \end{pmatrix}$$

Projective Transformation

$$H = \begin{pmatrix} h_{11} & h_{12} & h_{13} & h_{14} \\ h_{21} & h_{22} & h_{23} & h_{24} \\ h_{31} & h_{32} & h_{33} & h_{34} \\ h_{41} & h_{42} & h_{43} & h_{44} \end{pmatrix}$$

Invariant:

Orientation

Lengths

Angles

Parallelism

Collineation

Let x_1, x_2 and x_3 lie on a line I . Let H be a non-singular 3×3 matrix.
Then points Hx_i lie on the line $H^{-T}I$
Indeed,
 $H^{-T}I \cdot Hx_i = I^T H^{-1} Hx_i = 0$



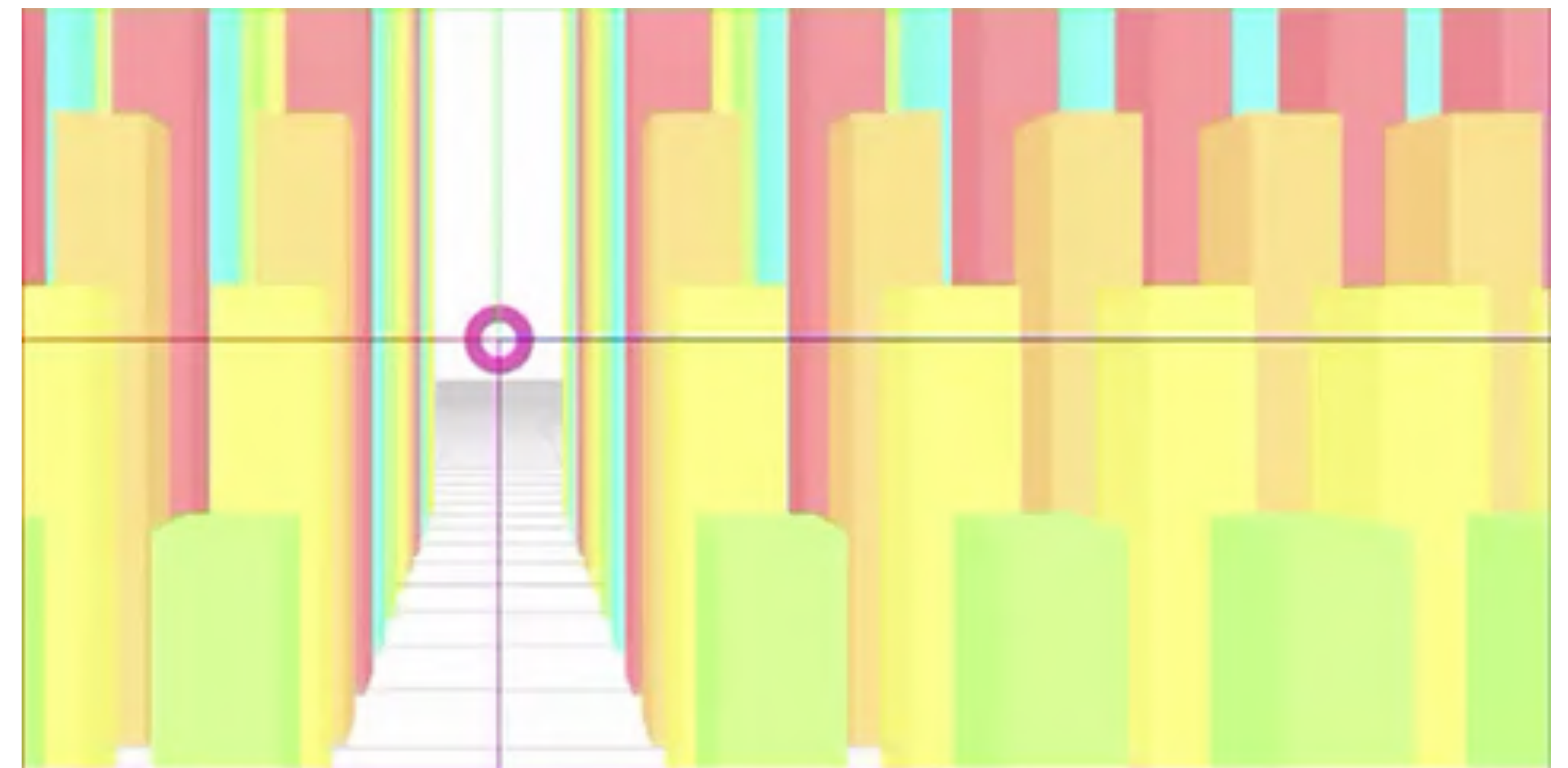
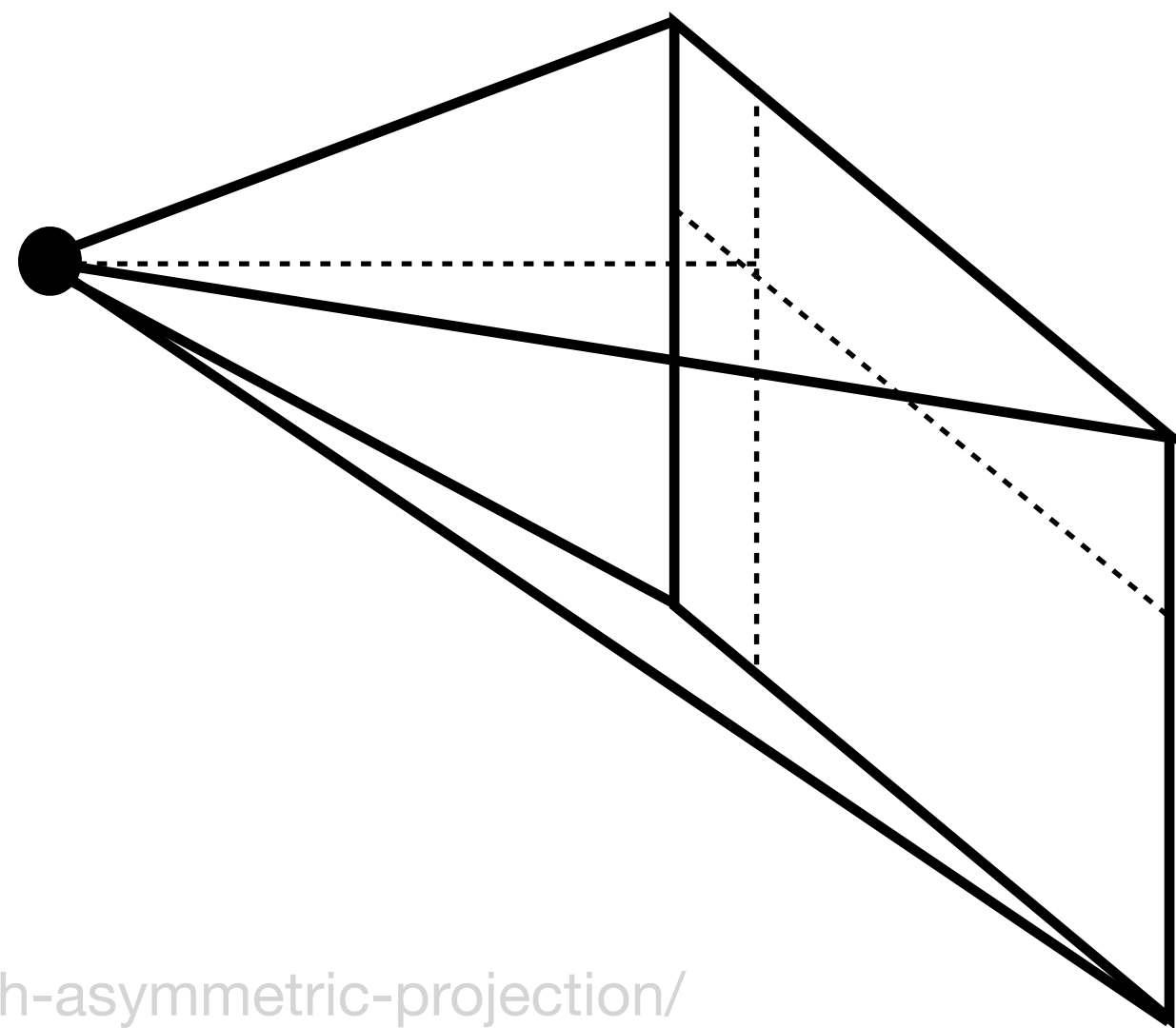


Camera Intrinsics / Projection Matrix

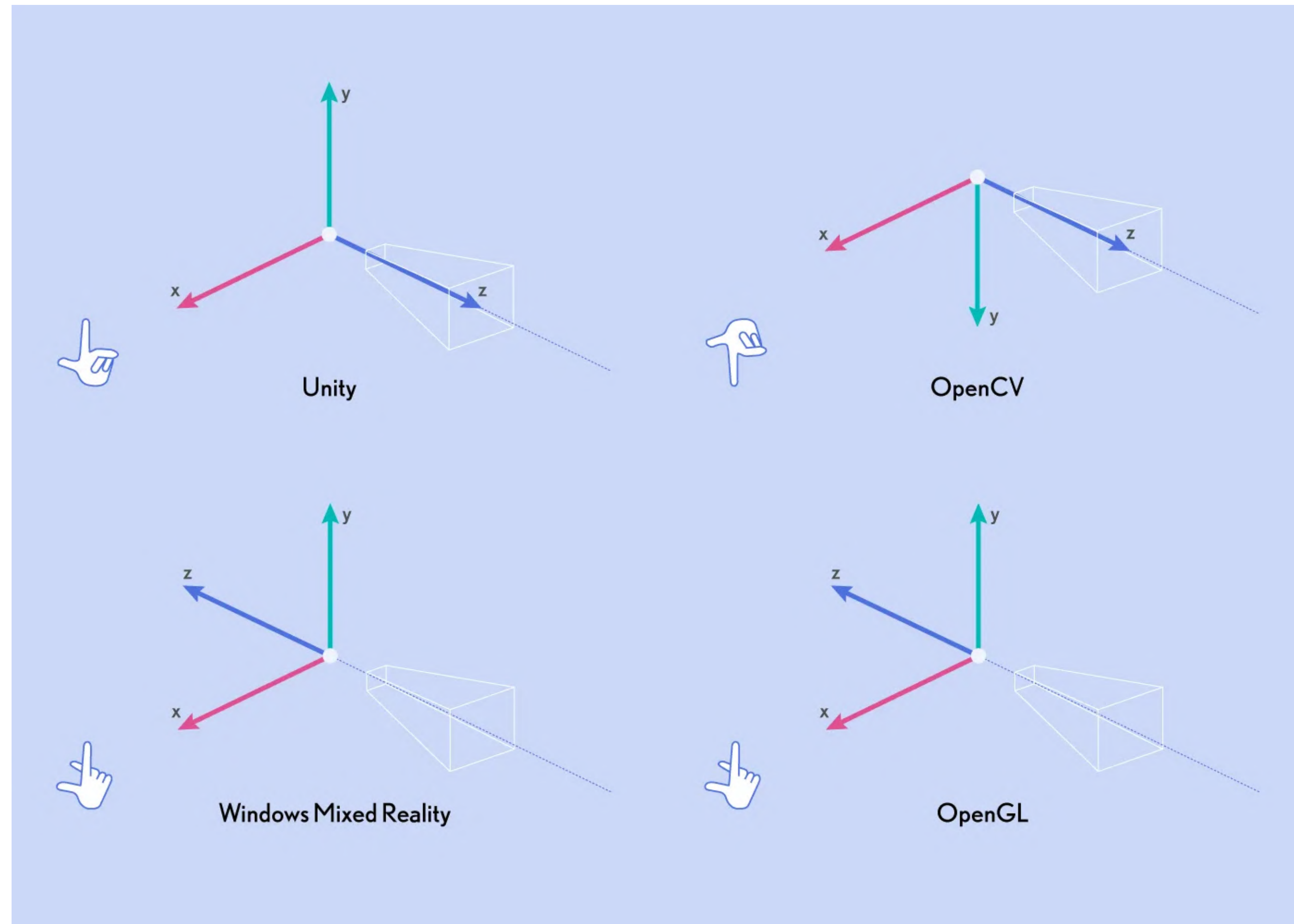
- There is one more component to the standard projection matrix

$$H = \begin{pmatrix} f_x & 0 & p_x & 0 \\ 0 & f_y & p_y & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} f_x & 0 & p_x \\ 0 & f_y & p_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} = K [I \mid 0]$$

- K represents *intrinsic* parameters of camera (“intrinsic calibration” = estimation of K)



A Note on Conventions



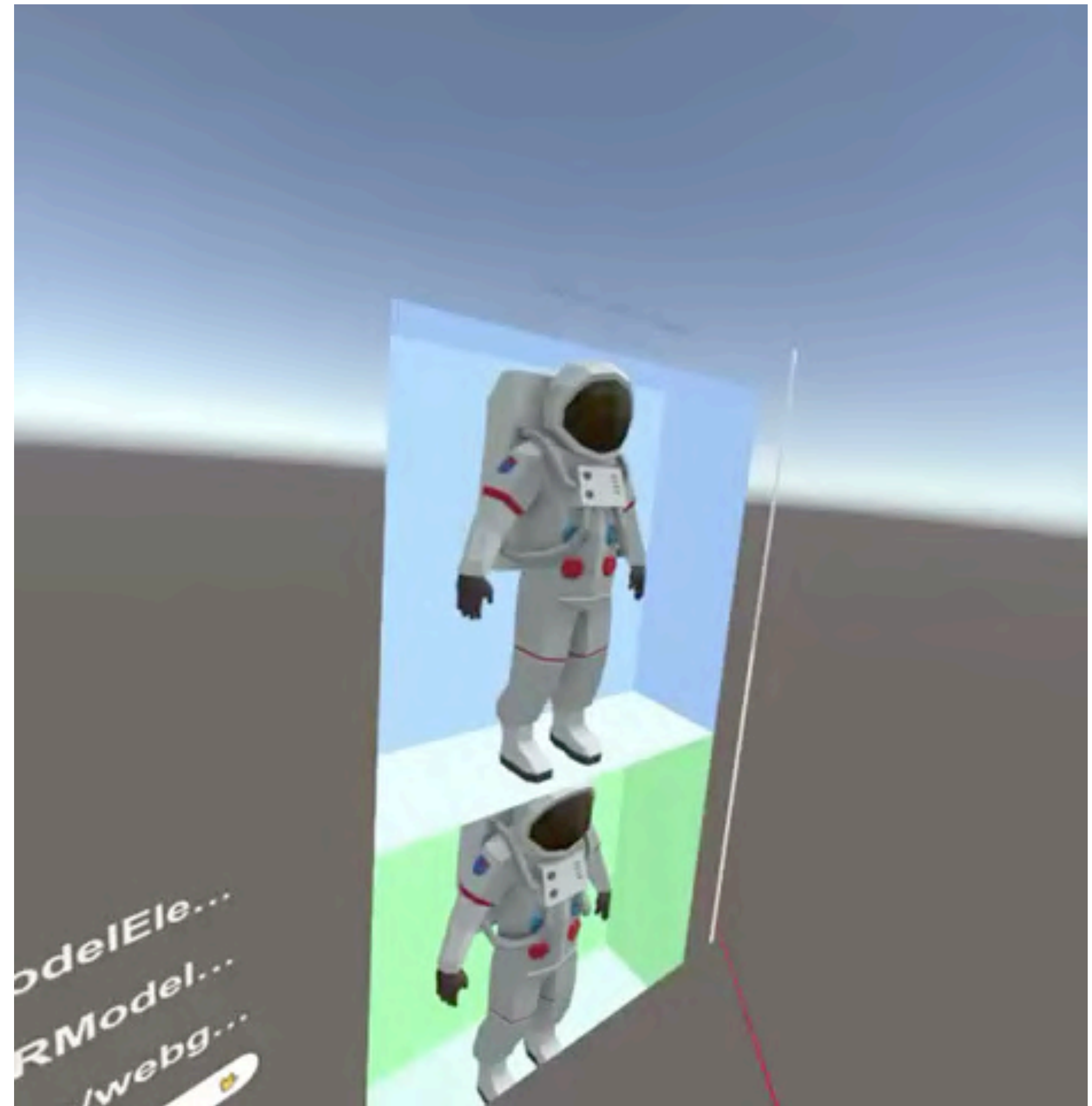
General Projections



General Projections



General Projections



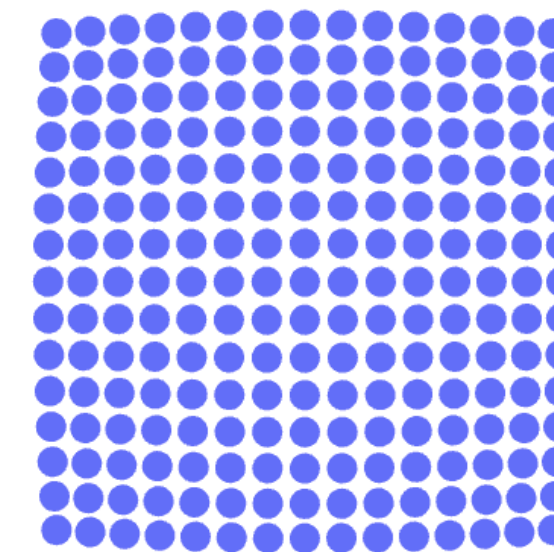
Model Limitations

Lens Distortions

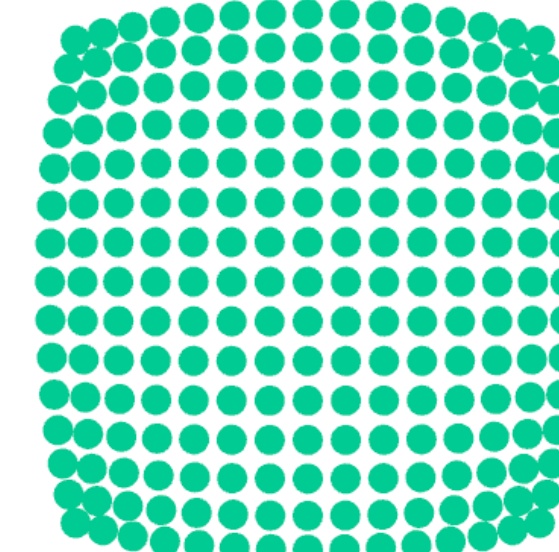


Use undistorted images!

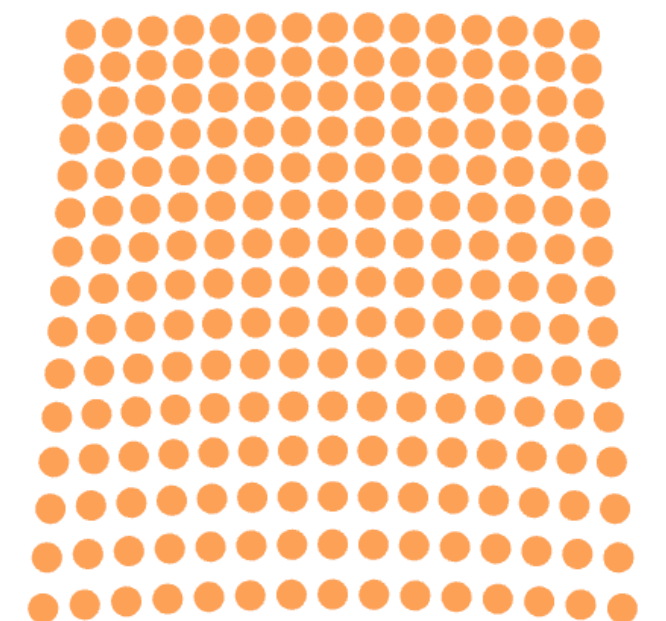
k1



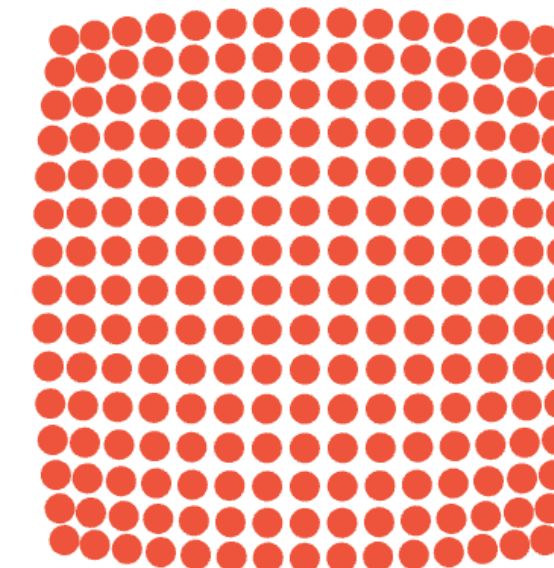
k2



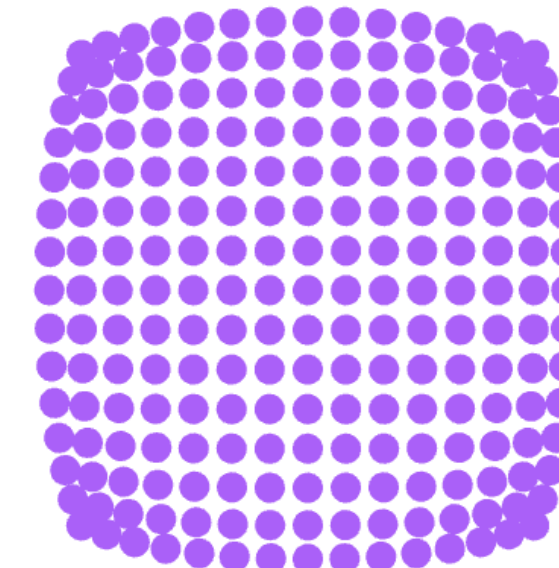
p1



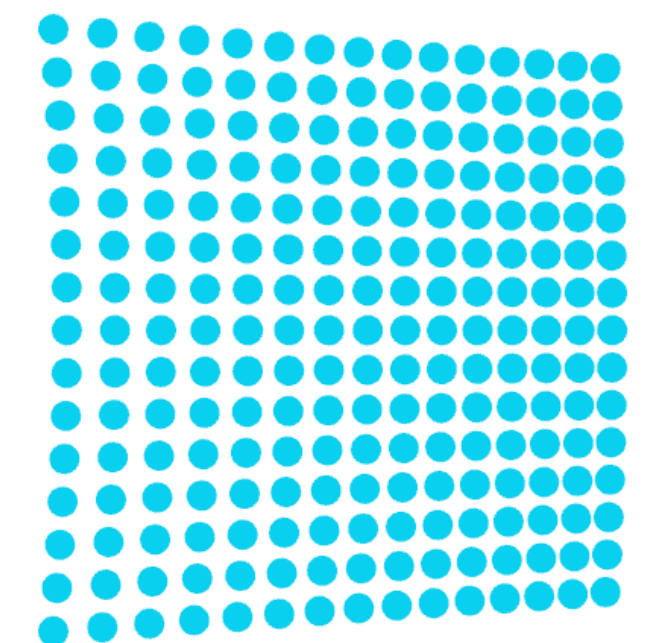
k3



k4

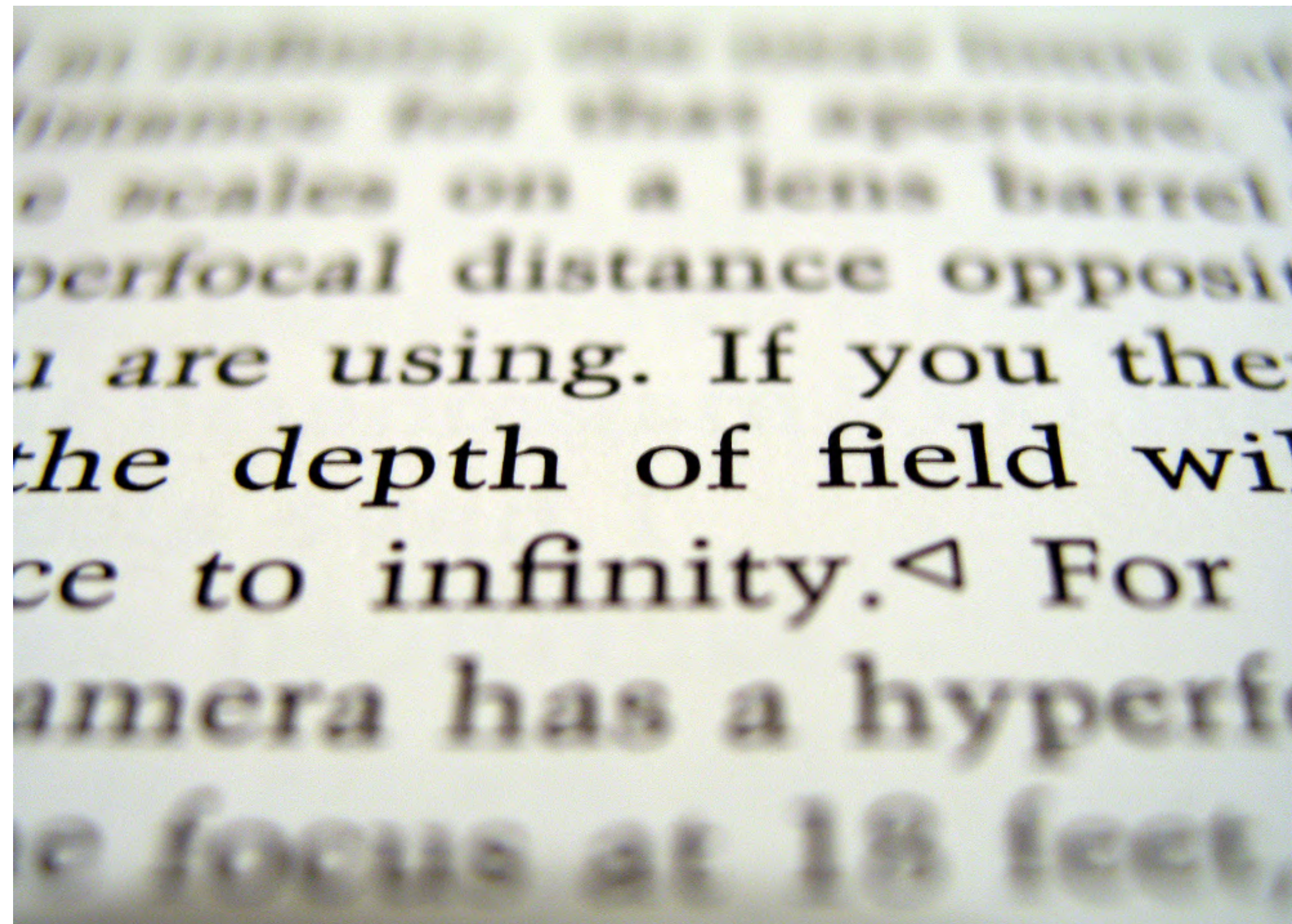


p2

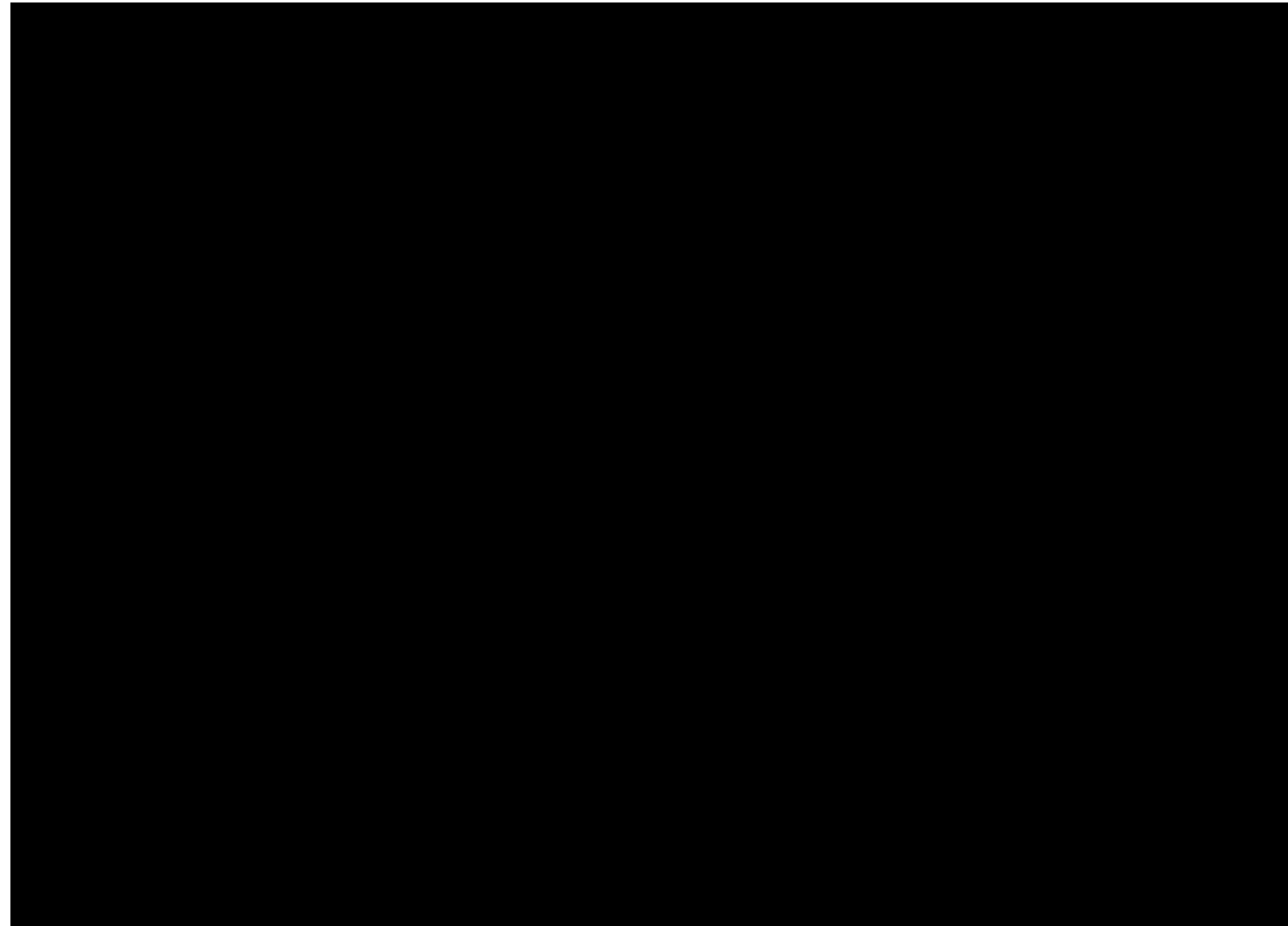
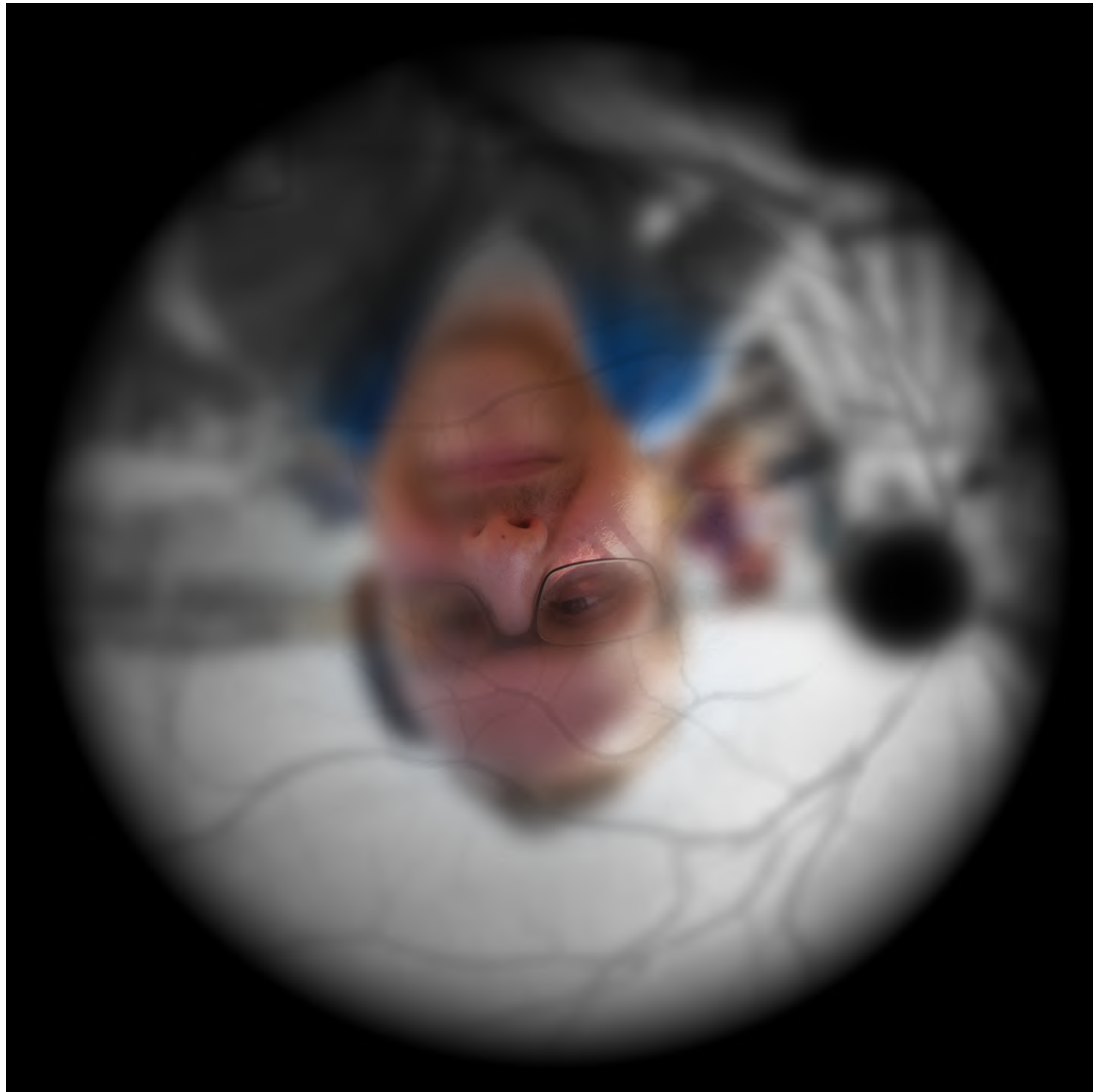


Model Limitations

No depth of field



What We Actually See



<http://ilab.usc.edu/retina/>

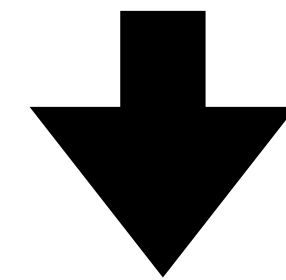
Digital Post-Processing Pipeline

Camera sensor measures luminance

Image files do not store actual luminance

Conversion into a file involves multiple steps:

- Demosaicking
- Tone mapping & White Balancing
- Denoising & Sharpening
- Compression



```
img = np.array([...], dtype=np.uint8)
```

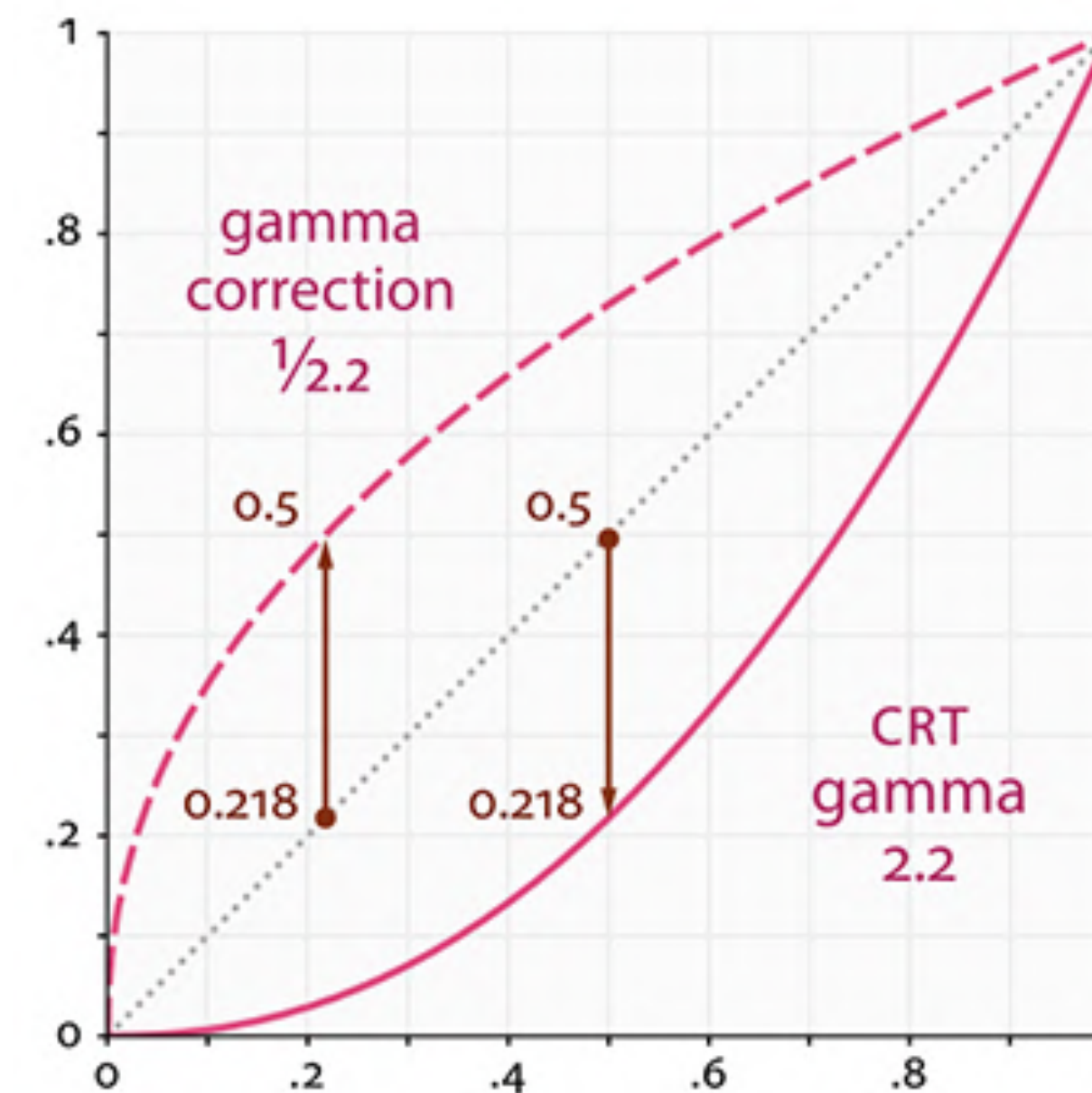
Gamma-Correction

- Our eyes are sensitive to relative luminance rather than absolute luminance
- In particular, we are more sensitive to luminance variations in darker regions rather than in bright ones

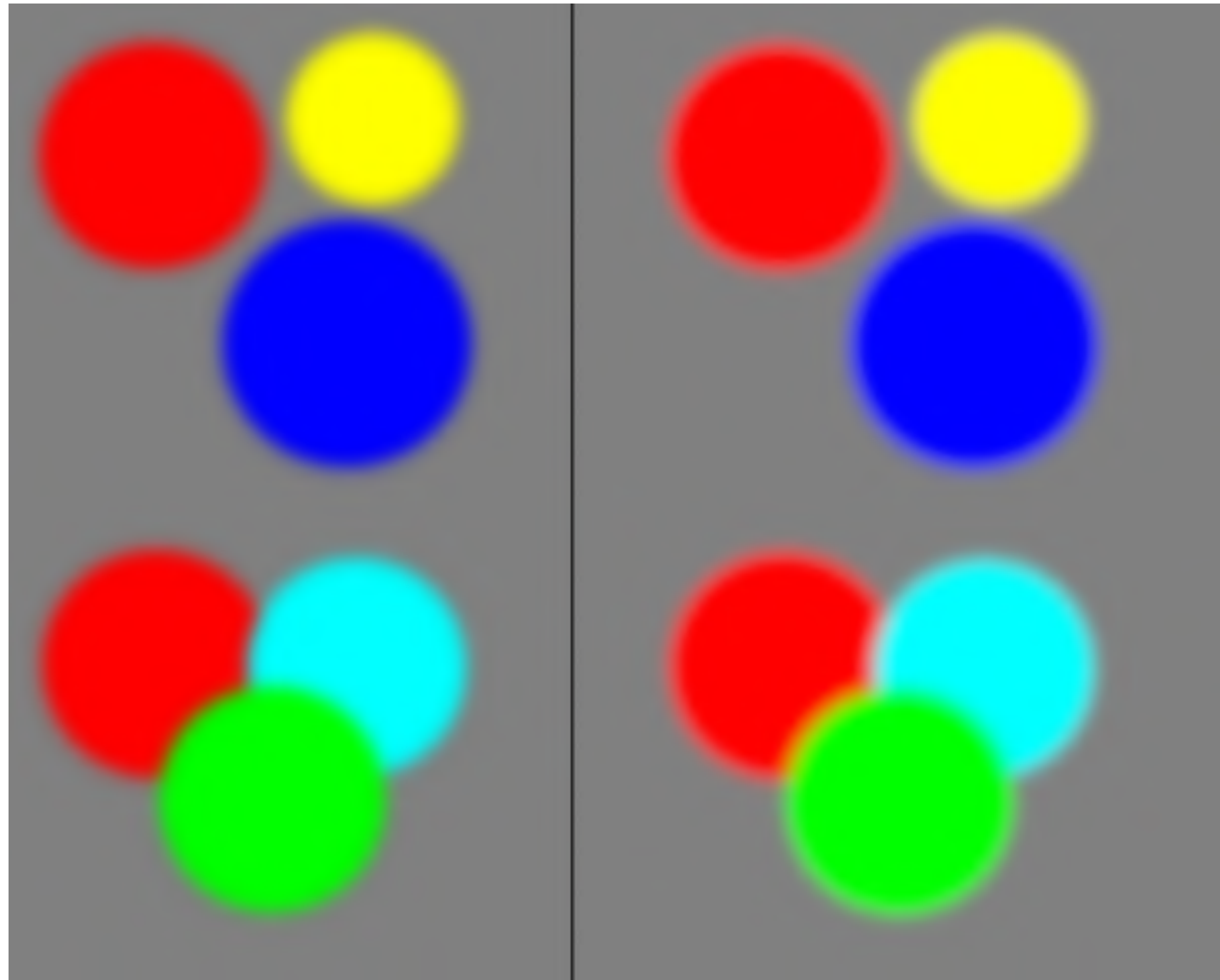


Gamma-Correction

- Gamma correction $f(x) = x^\gamma$ maps luminance so that darker regions are discretised more efficiently
- Encode images with $\gamma = \frac{1}{2.2}$ before discretisation
- JPG files are encoded (sRGB color space)
- EXR files to store uncompressed floats

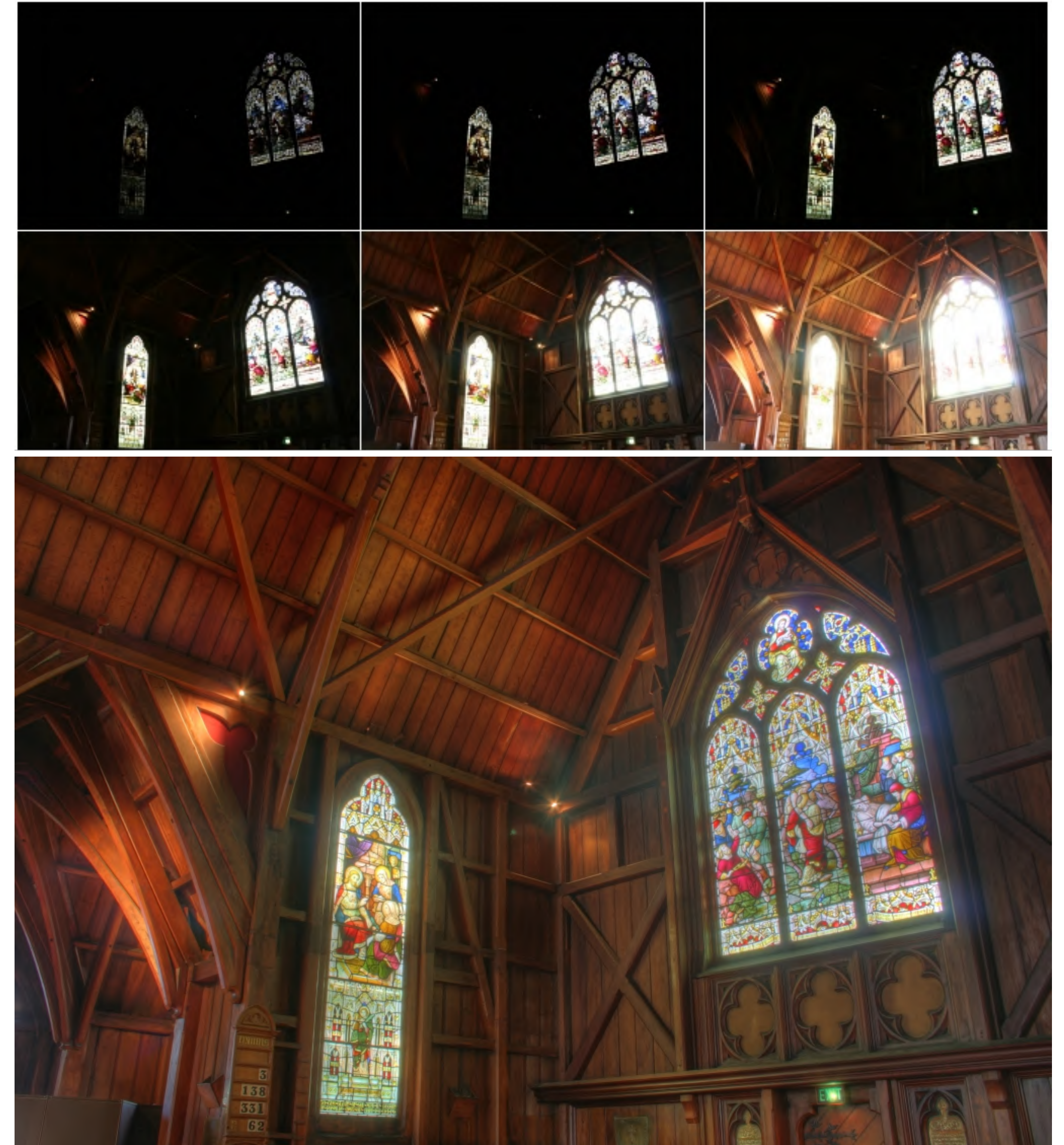


Operating in sRGB vs Linear Space



Blur encoded sRGB

Blur luminance



Locating the Camera

Camera Position

Extrinsic Parameters

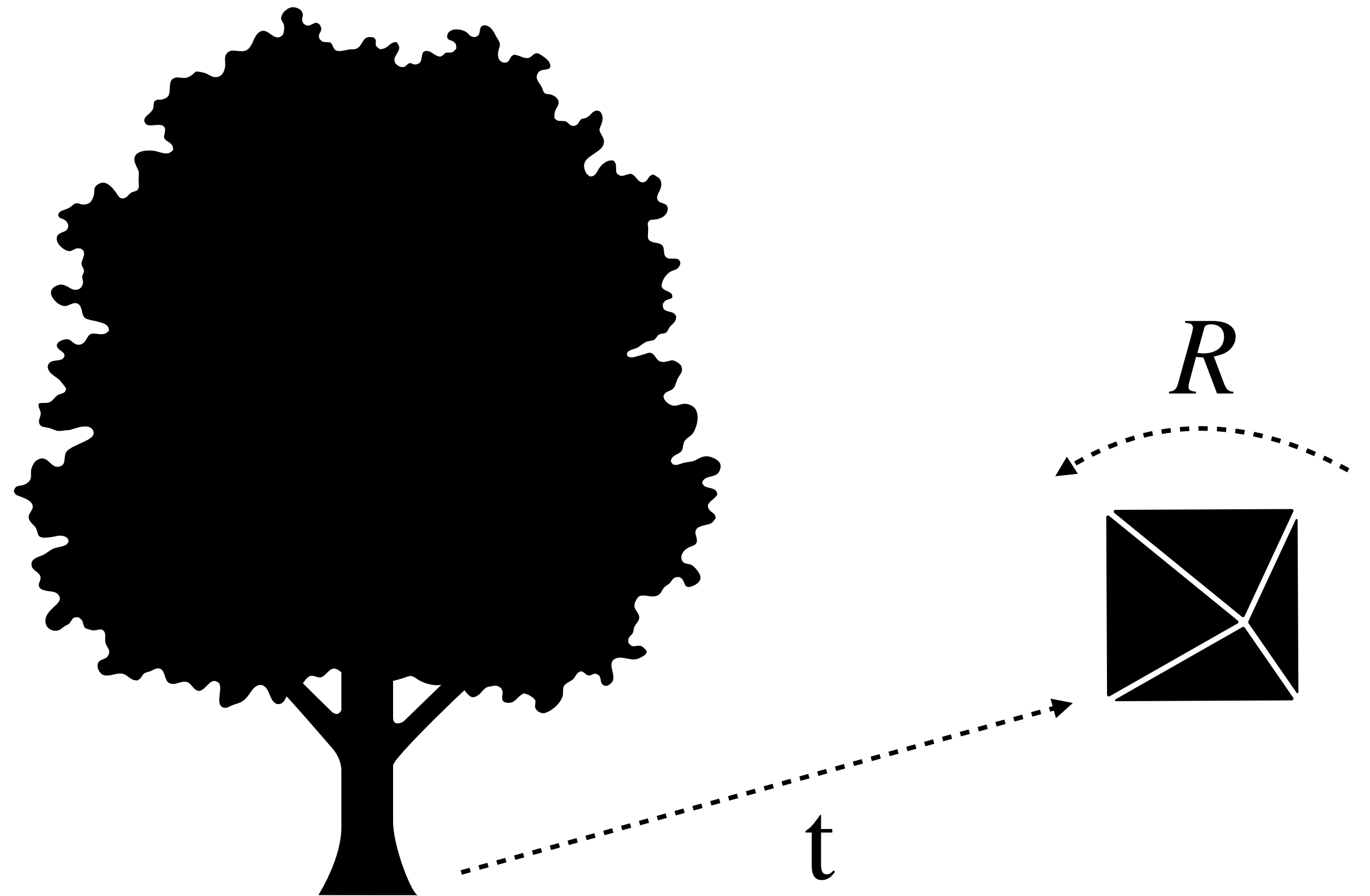
- Rigid movement $y = R(x - t)$
- In homogeneous coordinates translates into

$$y = \begin{bmatrix} R & -Rt \\ 0 & 1 \end{bmatrix} x = C^{W2C} x$$

- Perspective transformation turns into

$$H = \begin{bmatrix} f_x & 0 & p_x & 0 \\ 0 & f_y & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & -Rt \\ 0 & 1 \end{bmatrix}$$

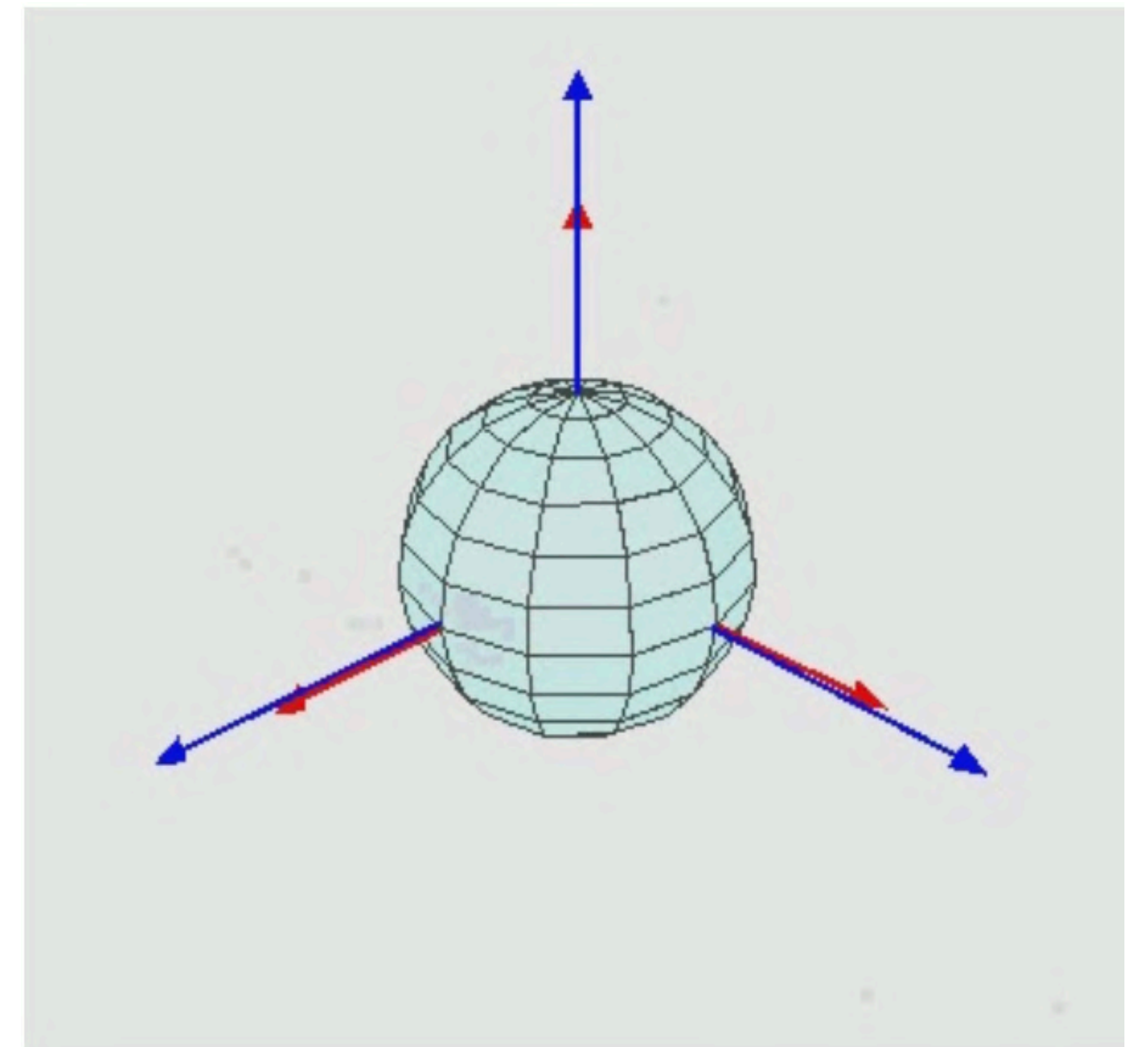
Degrees of freedom?



Rotation Representations

Euler Angles

- Composition of rotations along $z - x - z$ axes
- Can represent any rotation
- Is not injective
 - Loss of degree of freedom, gimbal lock



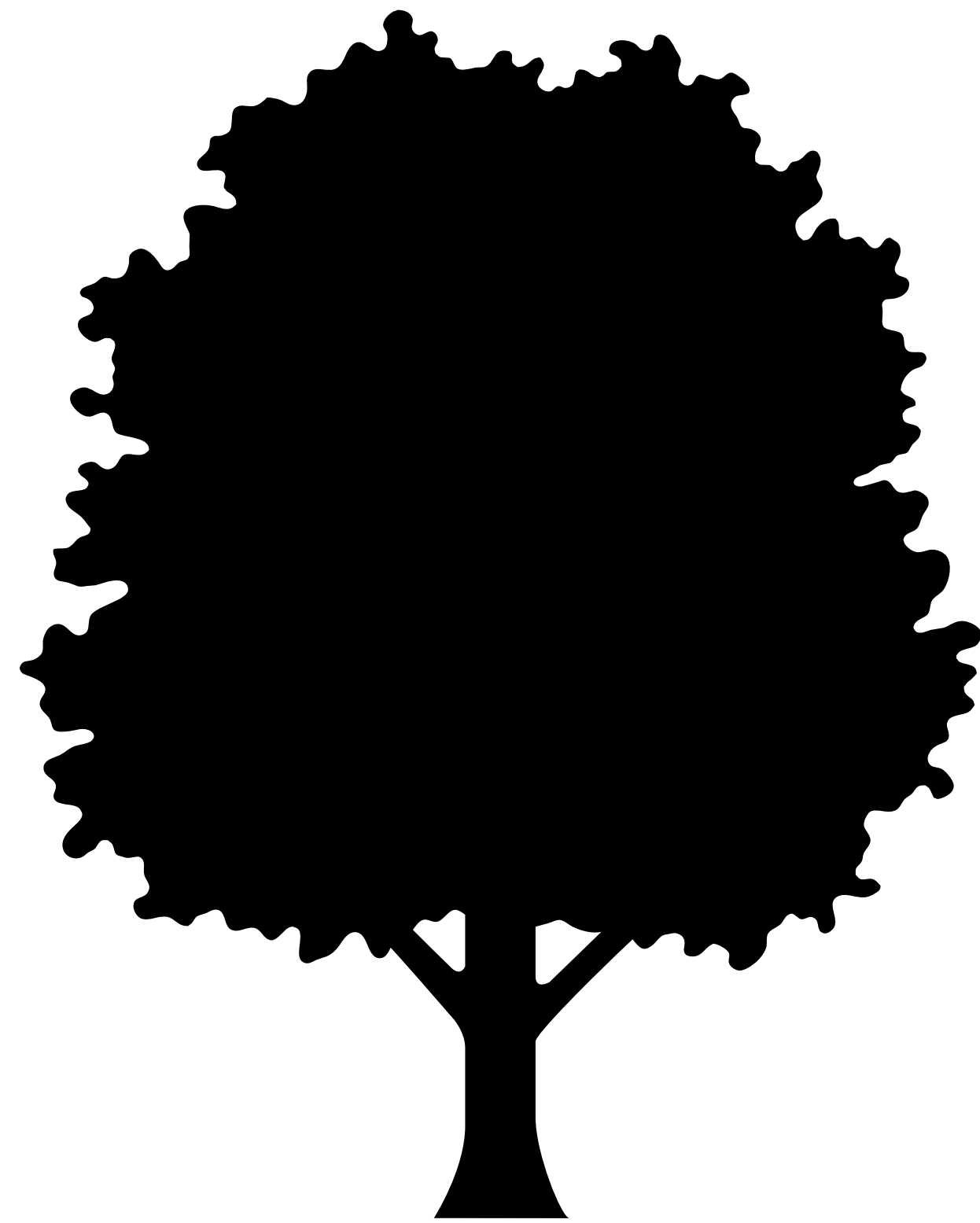
Rotation Representations

Angle Axis Representation

- Consider rotation $R \in SO(3)$
 - Eigenvalues lie on a unit circle in $\mathbb{C}$
 - $\|x\|^2 = (Rx, Rx) = \lambda^2 \|x\|^2 \Rightarrow \lambda^2 = 1$
 - Eigenvector n with $\lambda = 1$ defines rotation axis
 - Rotation angle θ
- Unit quaternions also represent $SO(3)$: $(n \cdot \sin \frac{\theta}{2}, \cos \frac{\theta}{2})$

“Look At” Camera

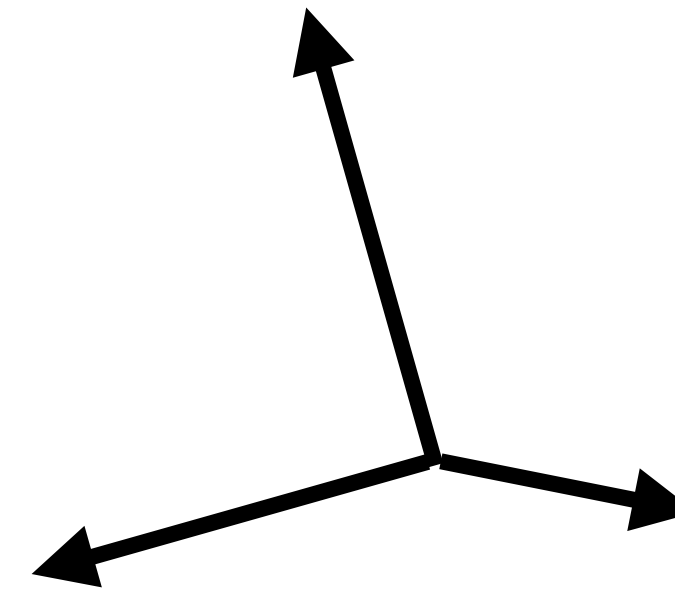
Looking at o from c



o

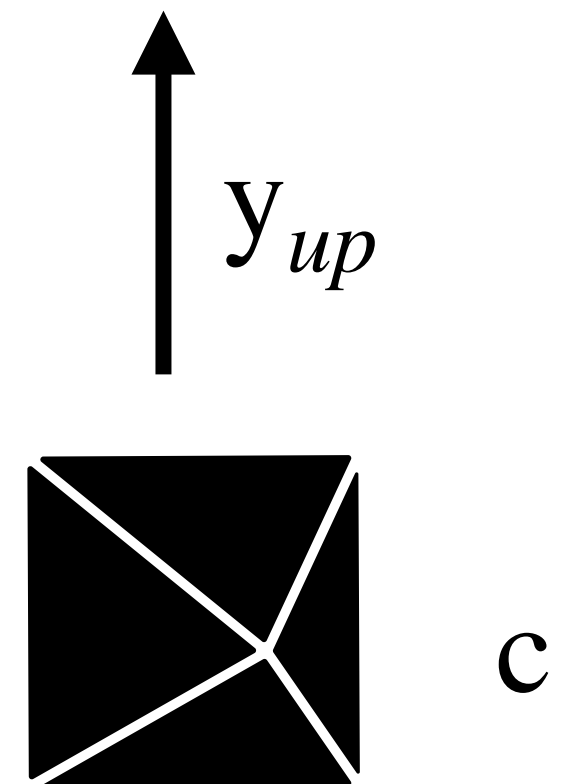
$$R = [x^T y^T z^T]$$

$$y = x \times z$$

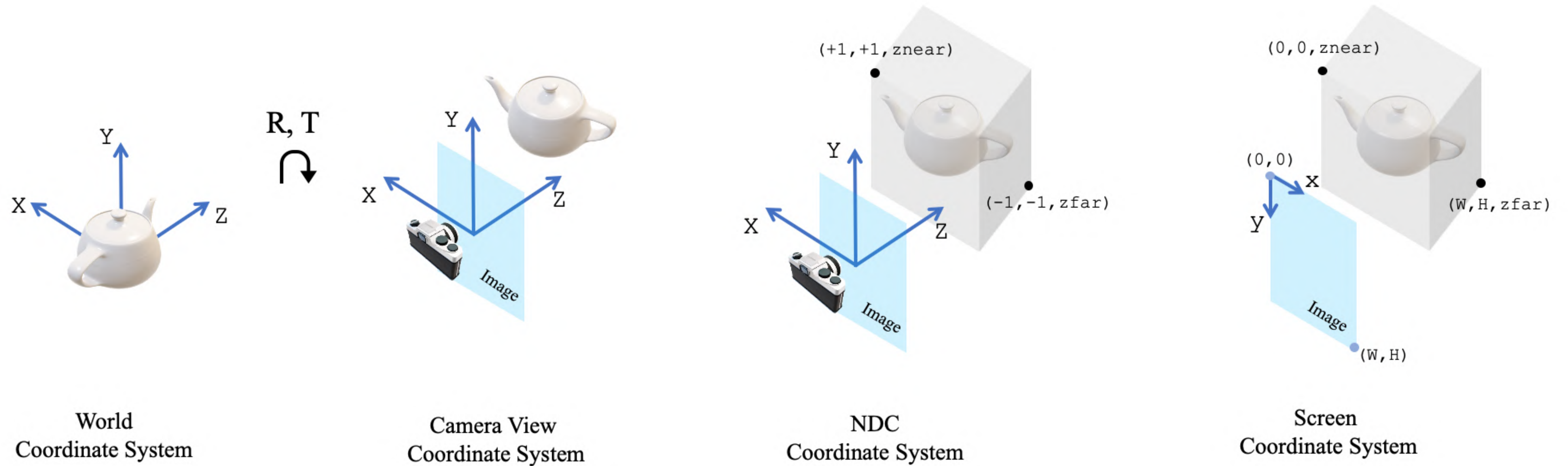


$$x = z \times y_{up}$$

$$z = \|o - c\|^{-1}(o - c)$$



Another Note on Conventions



Key Takeaways

- Pinhole camera models
- Perspective projections and homogeneous coordinates
- Extrinsic and intrinsic parameters of a camera
- Next lecture: multi-view geometry